Algebraic Hiddenness in the Sedenion Algebra: Characterization, Spectral Structure, and Bundle Topology

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Abstract

The 16-dimensional sedenion algebra $\mathbb{S}$, obtained as the fourth Cayley–Dickson doubling of the reals, is the lowest level of the CD tower at which the composition law fails and zero divisors appear. We develop a complete algebraic and geometric theory of its zero-divisor locus and the rank-4 vector bundle naturally associated with it.

The principal results are geometric. (i) The *Bundle Nontriviality Theorem* — the rank-4 kernel bundle over $G_2/\mathrm{SU}(2)$ carries a nonzero $\mathbb{Z}/2$ secondary characteristic class (the *aphanic twist*), and parallel transport around certain geodesic loops is $-\mathrm{id}$. (ii) The *Holonomy Decomposition Theorem* — given the previously established holonomy $U(2) = (U(1) \times \mathrm{SU}(2))/\mathbb{Z}_2$ of the canonical connection, the $\mathrm{SU}(2)$ factor is the restriction to the kernel of the lifted G_2-stabilizer action, and the $U(1)$ factor is generated, at the basepoint, by a commutator of right-multiplications R_{e_1}, R_{e_2}. (iii) The R_ℓ *Centrality Theorem* — that $U(1)$ generator is the restriction to the kernel of a single global ambient operator R_ℓ on $\mathbb{S}$, with $[R_\ell, L_p^\top L_p] = 0$ on the entire imaginary locus, of which the aphanic locus is a strict subset. (iv) The *CD-Tower Recursion Theorem* — that centrality identity is level-uniform, holding at every Cayley–Dickson level by an argument using only flexibility, inner-product duality, and the conjugation anti-homomorphism property.

These rest on an algebraic foundation which we develop by a self-contained route, and whose structural conclusions are in substantial part already known. (v) A *Pfaffian route* to the annihilator structure of $\mathbb{S}$: the identity $\mathrm{Pf}(L_p) = -(\|p\|^2)^2 T$, with $T = 4\|a\|^2\Re(b)^2 + 4\langle a, b\rangle^2 + (\|a\|^2 - \|b\|^2)^2$, certified by exact finite computation, from which follow in one derivation the zero-divisor criterion of Moreno and Khalil–Yiu, the $\{0, 4\}$ nullity dichotomy of Moreno and Biss–Dugger–Isaksen, and the determinant factorization $\det L_p = (\|p\|^2)^4 T^2$ obtained independently by Koebisu. (vi) A derivation of the *4-8-4 spectral structure* of the Gram operator $G_p = L_p^\top L_p$ — $\mathrm{Spec}(G_p) = \{(\|p\|^2 - \sigma)^4, (\|p\|^2)^8, (\|p\|^2 + \sigma)^4\}$, $\sigma = 2\|x \wedge y\|$ — by an associator-norm argument; this is equivalent to the eigenvalue computation of Biss–Christensen–Dugger–Isaksen for their operator $M_a = (1/\|a\|^2) L_{\bar{a}} L_a$, and is used here as the structural input to the bundle and holonomy results above. (vii) A unified eleven-step proof of the *Aphanic Characterization Theorem*, the doubly-imaginary case of the criterion in (v). All proofs are constructive and certified by self-contained C++17 probes.

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1 Introduction

1.1 The Cayley–Dickson Tower and Its Fourth Doubling

The Cayley–Dickson construction generates an infinite tower of real $*$-algebras

$$\mathbb{R} \subset \mathbb{C} \subset \mathbb{H} \subset \mathbb{O} \subset \mathbb{S} \subset \mathbb{P} \subset \mathbb{C}\widetilde{\otimes} \subset \cdots$$

by recursive application of the doubling formula

$$(a, b)(c, d) \;=\; (ac - \bar{d}b,\; da + b\bar{c}). \tag{1}$$

The first three doublings produce $\mathbb{C}$, $\mathbb{H}$, and $\mathbb{O}$. By Hurwitz's theorem (Baez, 2002; Hurwitz, 1898), no fourth normed real division algebra exists; the sedenion algebra $\mathbb{S}$, the fourth doubling, is no longer a division algebra. The composition law $\|ab\| = \|a\|\,\|b\|$ fails in $\mathbb{S}$, and zero divisors appear: there exist $p, q \in \mathbb{S} \setminus \{0\}$ with $pq = 0$. The locus of left zero divisors is the focus of this paper.

The standard view of this transition is degradative. The associative, normed, and division properties are progressively shed at each doubling, and beyond $\mathbb{O}$ what remains is regarded as an algebraic curiosity rather than a structural object of independent interest. We take a different view. The zero-divisor locus of $\mathbb{S}$ is a smooth real submanifold with a G_2-symmetry, the kernel of the multiplication operator L_p has constant dimension 4 on this locus, and the family of kernels assembles into a topologically nontrivial vector bundle whose canonical connection has holonomy in the unitary group $U(2)$ (established previously; see Section 7, where that identification is taken as given and its factors are located inside $\mathfrak{so}(\mathcal{K})$). What from one perspective is a degeneracy, from another is a precisely defined geometric object.

1.2 Setting and Conventions

We work throughout with the canonical recursive Cayley–Dickson multiplication (1). Hand-written multiplication tables based on Fano-plane bookkeeping are not equivalent to the recursive formula in detail (sign conventions differ in 24 of the $2 \times 8 \times 7 = 112$ products that are nonzero), and the choice matters: which pairs (x, y) produce a zero divisor depends on the sign convention. The recursive product is the canonical choice in the modern non-associative algebra literature (Baez, 2002; Schafer, 1966) and is the convention used throughout. All numerical certifications referenced in this paper are run with this convention.

For $a \in \mathbb{S}$, $L_a, R_a : \mathbb{S} \to \mathbb{S}$ denote the 16×16 real matrices of left and right multiplication by a. We write $p = (x, y) \in \mathbb{S}$ for $p = \iota_1(x) + \iota_2(y)$ where ι_1, ι_2 are the canonical inclusions of $\mathbb{O}$ into the two halves of $\mathbb{S}$. The inner product on $\mathbb{S}$ is the natural one inherited from $\mathbb{O}$, and the imaginary subspace is $\mathrm{Im}(\mathbb{S}) = \mathrm{Im}(\mathbb{O}) \oplus \mathbb{O} \cdot \ell$.

1.3 Principal Results

1. **Section 3 (Characterization).** A unified algebraic proof that $\dim \ker L_p > 0$ if and only if $\|x\| = \|y\|$ and $\langle x, y\rangle = 0$, with kernel dimension exactly 4 in that case (Theorem 3.1). The proof passes through the keystone identity $(\|x\|^2 - \|y\|^2)^2 + 4\langle x, y\rangle^2 = 0$ whose non-negative summands force both conditions simultaneously.

2. **Section 4 (4-8-4 Spectrum).** The Gram operator $G_p = L_p^\top L_p$ has spectrum $\{(\|p\|^2 - \sigma)^4, (\|p\|^2)^8, (\|p\|^2 + \sigma)^4\}$ with $\sigma = 2\|x \wedge y\|$ (Theorem 4.1). The aphanic locus is exactly where the lowest block hits 0, and the iconic "4-8-4" multiplicities arise from the Cayley–Dickson block decomposition combined with an isometry-difference structure of the off-diagonal block.

3. **Section 5 (Annihilator Fourfold Theorem).** The full left-annihilator dichotomy for an *arbitrary* nonzero sedenion, dropping the standing assumption that the second component is imaginary (Theorem 5.14): a nonzero real part forces a trivial kernel; for imaginary $p = (a, b)$ with $b \in \mathbb{O}$ arbitrary, $\ker L_p \neq 0$ if and only if $\Re(b) = 0$, $\langle a, b\rangle = 0$, and $|a| = |b|$, with $\dim_{\mathbb{R}} \ker L_p$ exactly 0 or 4 and $\dim_{\mathbb{R}} \ker L_p = \dim_{\mathbb{R}} \ker R_p$. The mechanism is the determinant identity $\det L_p = (|p|^2)^4 T^2$ with $T = 4|a|^2\Re(b)^2 + 4\langle a, b\rangle^2 + (|a|^2 - |b|^2)^2$, whose three non-negative summands force the criterion; the value 4 is the multiplicity count of Theorem 3.1.

4. **Section 6 (Bundle Topology).** The aphanic locus $\mathcal{V} \cong G_2/\mathrm{SU}(2)$ is an 11-dimensional homogeneous space, and the kernel map $\Phi : \mathcal{V} \to \mathrm{Gr}(4, \mathbb{S})$ realizes a rank-4 real bundle $\mathcal{K} \to \mathcal{V}$ with $\mathrm{Aut}(\mathbb{O})$-equivariance. We show that $\mathcal{K}$ carries a nonzero secondary $\mathbb{Z}/2$ characteristic class (Theorem 6.4), and the canonical connection has holonomy $-\mathrm{id}$ around certain geodesic loops (Theorem 6.6).

5. **Section 7 (Holonomy Decomposition).** The full holonomy algebra at the basepoint decomposes algebraically as $\mathfrak{u}(2) = \mathfrak{u}(1)_R \oplus \mathfrak{su}(2)_L$, where the $\mathfrak{su}(2)$ factor is the restriction to $\mathcal{K}$ of the lifted action of the $\mathrm{SU}(2) \subset G_2$ stabilizer of the basepoint, and the $\mathfrak{u}(1)$ factor is generated by a commutator of right-multiplications by orthogonal quaternionic units (Theorem 7.2).

6. **Section 8 (R_ℓ Centrality).** The $\mathfrak{u}(1)$ generator at the basepoint is, up to scale, the restriction to $\mathcal{K}$ of a single ambient operator R_ℓ on $\mathbb{S}$, and $[R_\ell, G_p] = 0$ on the entire imaginary locus (Theorem 8.1). The aphanic conditions $\|x\| = \|y\|$ and $\langle x, y\rangle = 0$ are not used in the proof.

7. **Section 9 (Tower Recursion).** The centrality identity is the $\mathbb{S}$-level instance of a tower-wide recursion that propagates via the *Imaginary L–R Symmetry Lemma* $L_{\bar{X}} L_X = R_{\bar{X}} R_X$ on $\mathrm{Im}(\mathcal{A}_n)$, a flexibility corollary preserved at every CD level (Theorem 9.1). The pathion-level analogue is proven algebraically (Theorem 9.3), and empirical confirmation is given at the chingon level.

8. **Section 10 (Open Frontiers).** A summary of the questions left open: persistence of the holonomy classification at higher CD levels, the algebraic identification of the SU(2) factor with right-multiplication by Fano units, and the global integrability of the kernel distribution.

1.4 Computational Reproducibility

Every algebraic claim in the paper is accompanied by a self-contained C++17 probe that verifies the claim numerically at machine precision. The probes are released as a single archive (`the_aphanic_toy`) with no external dependencies, building under both MSVC and GCC. Where proofs are algebraic, the probes serve as referee-checkable witnesses; where proofs are empirical (e.g., dim 64 chingon recursion), the probes are the primary evidence. All residuals reported in the paper are reproducible to within IEEE double rounding.

1.5 Relationship to the Companion Paper

The framework paper (Skiouris, 2026) formalizes the *Aphanics* cognitive monitoring framework at the abstract level: the kernel of an internal action operator, the curvature of its variation, and the topological winding number of closed dialectical arcs. That paper is a framework proposal whose structural credibility depends on the existence of at least one concrete algebraic setting in which the abstract axioms are realized. The present paper supplies that setting — the sedenion algebra — with full proofs of the realization. The two papers are independent in their formal

content but mutually reinforcing: the abstract framework explains why the specific algebraic results matter, and the specific algebraic results demonstrate that the framework is not vacuous.

2 Setup and Conventions

2.1 The Cayley–Dickson Tower

We adopt the notation $\mathcal{A}_n$ for the nth Cayley–Dickson algebra, with $\mathcal{A}_0 = \mathbb{R}$ and $\mathcal{A}_{n+1} = \mathcal{A}_n \oplus \mathcal{A}_n \cdot \ell_{n+1}$ where $\ell_{n+1}^2 = -1$ and ℓ_{n+1} anticommutes with every imaginary element of $\mathcal{A}_n$. Multiplication is given by recursive application of (1). The first six levels are

$$\mathcal{A}_0 = \mathbb{R}, \quad \mathcal{A}_1 = \mathbb{C}, \quad \mathcal{A}_2 = \mathbb{H}, \quad \mathcal{A}_3 = \mathbb{O}, \quad \mathcal{A}_4 = \mathbb{S}, \quad \mathcal{A}_5 = \mathbb{P}, \quad \mathcal{A}_6 = \mathbb{C}\approx.$$

We write ℓ for $\ell_4 = e_8$, the doubling generator of $\mathbb{S}$ over $\mathbb{O}$, which is identified with the basis element with index 8 when $\mathbb{S}$ is given the standard ordered basis $\{1, e_1, \ldots, e_{15}\}$. Similarly $\ell_5 = e_{16}$ is the doubling generator of $\mathbb{P}$ over $\mathbb{S}$, and $\ell_6 = e_{32}$ that of $\mathbb{C}\approx$ over $\mathbb{P}$.

2.2 Properties Surviving Each Doubling

The properties enjoyed by $\mathcal{A}_n$ degrade at each doubling. The relevant landmarks are recorded in Table 1.

Table 1: Algebraic properties surviving each Cayley–Dickson doubling.

Property	$\mathbb{R}$	$\mathbb{C}$	$\mathbb{H}$	$\mathbb{O}$	$\mathbb{S}$	$\mathbb{P}$	higher
Commutative	yes	yes	no	no	no	no	no
Associative	yes	yes	yes	no	no	no	no
Alternative	yes	yes	yes	yes	no	no	no
Composition	yes	yes	yes	yes	no	no	no
Division algebra	yes	yes	yes	yes	no	no	no
Flexible	yes	yes	yes	yes	yes	yes	yes
Conjugation $*$-anti-hom.	yes	yes	yes	yes	yes	yes	yes

The key observation underpinning the present work is that the two properties surviving every doubling — flexibility $a(ba) = (ab)a$ and the conjugation anti-homomorphism $\overline{ab} = \bar{b}\bar{a}$ — are sufficient to drive the R_ℓ centrality identity (Section 8) and its tower-wide recursion (Section 9). The characterization theorem of Section 3 relies on the stronger composition and alternativity properties, which is why the dim-16 case is the lowest level at which a nontrivial kernel exists.

2.3 Standing Assumptions

Unless stated otherwise:

- $p = (x, y) \in \mathbb{S}$ with $x, y \in \mathrm{Im}(\mathbb{O})$ (the *imaginary locus*).
- $L_p, R_p \in \mathrm{End}(\mathbb{S})$ are the matrices of left and right multiplication, in the standard ordered basis.
- $G_p = L_p^\top L_p$ is the Gram operator.
- $\sigma = 2\|x \wedge y\| = 2\sqrt{\|x\|^2\|y\|^2 - \langle x, y\rangle^2}$ is the wedge norm.
- $\widehat{\mathcal{V}} \subset \mathbb{S}$ is the *aphanic cone*, defined by $\|x\| = \|y\|$, $\langle x, y\rangle = 0$, and x, y linearly independent; it is invariant under positive scaling.

- $\mathcal{V} := \{p \in \widehat{\mathcal{V}} : \|x\| = \|y\| = 1\}$ is the *aphanic locus*, the unit-normalized slice of the cone (equivalently $\|p\|^2 = 2$). **Unless stated otherwise, $\mathcal{V}$ denotes this unit-normalized locus throughout.** The spectral statements are homogeneous in $\|p\|$ and are stated on the cone $\widehat{\mathcal{V}}$; the bundle and holonomy statements are stated on $\mathcal{V}$.
- $\mathcal{K} \to \mathcal{V}$ is the rank-4 real bundle $\mathcal{K}_p = \ker L_p$.

2.4 A Note on Empirical Certification

Throughout the paper, results are classified as PROVEN (algebraic proof, no appeal to numerics), EMPIRICAL (verified numerically at high precision but without a known algebraic proof), or CONFIRMED (stated by other authors, verified here). Each theorem statement records its status. Empirical residuals quoted are worst-case across the cited number of random samples.

These three words are reserved for claims. The computational witnesses carry a separate vocabulary: code reports PASS, FAIL, or CERTIFIED, and never PROVED or PROVEN. A certification must state its epistemic basis — either the numerical tolerance at which it was evaluated, or *exact*, when the result is established by exhaustive finite verification or by exact arithmetic. Tier vocabulary belongs to claims, witness vocabulary to computation, and the two are never interchanged.

The separation is not a concession. Where bilinearity reduces an identity to finitely many basis triples and every one is checked exactly, this manuscript may correctly say the identity is proved: the program reports what it did, exact certification, and the mathematics states what follows from it, a proof. Saying both, in their own words, is more precise than letting one word carry both senses.

2.5 Definitions

The following three terms are used throughout, including inside formal statements. Each is fixed here, before first such use.

Definition 2.1 (Aphanic element)**.** Let $p = (x, y) \in \mathbb{S}$ under the Cayley–Dickson identification $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}$. Then p is *aphanic* if $x, y \in \mathrm{Im}(\mathbb{O})$ and

$$\|x\| = \|y\| \neq 0, \qquad \langle x, y\rangle = 0,$$

where $\langle \cdot, \cdot \rangle$ and $\| \cdot \|$ are the standard Euclidean inner product and norm on $\mathbb{O} \cong \mathbb{R}^8$.

Definition 2.2 (Aphanic locus)**.** The *aphanic locus* is the set $\mathcal{A} := \{\, p \in \mathbb{S} : p \text{ is aphanic} \,\}$ of all elements satisfying Definition 2.1.

Definition 2.3 (Aphanic twist)**.** The *aphanic twist* of the kernel bundle $\mathcal{K} \to \mathcal{V}$ is its secondary characteristic class

$$[\mathcal{K}] \;\in\; H^6(\mathcal{V}; \mathbb{Z}/2) \;\cong\; \mathbb{Z}/2.$$

Remark 2.4. The name records what the definition does not: an aphanic p is a nonzero element of $\mathbb{S}$ annihilated, in the sense made precise in Section 3, by the very operator its own system uses to measure it. The two conditions of Definition 2.1 — equal norms and orthogonality — are exactly the conditions under which that annihilation occurs, and Definition 2.2 names the resulting set.

Definition 2.3 names an invariant, not a phenomenon. The aphanic twist exists as a class whatever its value; whether it vanishes is the content of Theorem 6.4, and the associated $-\mathrm{id}$ transport behaviour is treated separately where it is established. The definition deliberately does not identify the class with a holonomy character.

3 The Aphanic Characterization Theorem

This section gives an unbroken algebraic derivation of the central theorem of the framework: for $p = (x, y) \in \mathbb{S}$ with $x, y \in \mathrm{Im}(\mathbb{O})$, the left-multiplication operator L_p has a nontrivial kernel if and only if $\|x\| = \|y\|$ and $\langle x, y\rangle = 0$; and when these conditions hold, $\dim\ker L_p = 4$. The proof is an eleven-step chain from the Cayley–Dickson multiplication formula through composition-algebra identities, Artin's theorem, and the seven-dimensional associator norm formula, terminating in the single algebraic identity $(\|x\|^2 - \|y\|^2)^2 + 4\langle x, y\rangle^2 = 0$ whose non-negative summands force both conditions simultaneously.

3.1 Setup

The sedenion algebra is constructed by Cayley–Dickson doubling of the octonions, $\mathbb{S} = \mathbb{O} \oplus \mathbb{O} \cdot \ell$, with multiplication

$$(a, b)(c, d) \;=\; (ac - \bar{d}b,\; da + b\bar{c}), \qquad a, b, c, d \in \mathbb{O}. \tag{2}$$

Here $\bar{\cdot}$ denotes octonion conjugation: if $u = u_0 + u_{\mathrm{im}}$ with $u_0 \in \mathbb{R}$ and $u_{\mathrm{im}} \in \mathrm{Im}(\mathbb{O})$, then $\bar{u} = u_0 - u_{\mathrm{im}}$. The inner product on $\mathbb{O}$ is $\langle u, v\rangle = \frac{1}{2}(u\bar{v} + v\bar{u})_0$, and that on $\mathbb{S}$ extends as $\|(x, y)\|^2 = \|x\|^2 + \|y\|^2$.

For $a \in \mathbb{O}$, denote by $L_a, R_a : \mathbb{O} \to \mathbb{O}$ the 8×8 real matrices representing left and right multiplication by a. Let C denote the 8×8 conjugation matrix. Three identities will be used repeatedly:

1. *Composition law:* $L_a L_{\bar{a}} = R_a R_{\bar{a}} = \|a\|^2 I$ for all $a \in \mathbb{O}$.
2. *Skew-adjointness on imaginaries:* for $a \in \mathrm{Im}(\mathbb{O})$, $L_a^T = -L_a$ and $R_a^T = -R_a$.
3. *Alternativity:* $\mathbb{O}$ is alternative, so the associator $[u, v, w] := (uv)w - u(vw)$ is totally antisymmetric in its three arguments on $\mathrm{Im}(\mathbb{O})$.

In particular, for $a \in \mathrm{Im}(\mathbb{O})$ we have $\bar{a} = -a$, so the composition law specialises to $L_a^2 = -\|a\|^2 I$ and $R_a^2 = -\|a\|^2 I$.

Theorem 3.1 (Aphanic Characterization)**.** *Let $p = (x, y) \in \mathbb{S}$ with $x, y \in \mathrm{Im}(\mathbb{O})$. Then*

$$\ker L_p \neq \{0\} \iff \|x\| = \|y\| \;\; \textit{and} \;\; \langle x, y\rangle = 0. \tag{3}$$

When these conditions hold, $\dim\ker L_p = 4$.

3.2 The Proof

The proof proceeds in eleven steps. Each step is a theorem of elementary octonion algebra; none appeals to numerical evidence.

Step 1 (Cayley–Dickson Block Structure). Applying (2) to $p = (x, y)$ acting on $z = (u, v)$ gives $p \cdot z = (xu - \bar{v}y,\; vx + y\bar{u})$. For $x, y \in \mathrm{Im}(\mathbb{O})$ the operator on $(u, v)^T$ is the 16×16 block matrix

$$L_p \;=\; \begin{pmatrix} L_x & -R_y C \\ L_y C & R_x \end{pmatrix}. \tag{4}$$

Here $\bar{v}y = R_y C(v)$ and $y\bar{u} = L_y C(u)$; the sign on $R_y C$ comes from (2) directly.

Step 2 (Diagonal Blocks of the Gram Operator). Let $G := L_p^T L_p$. Using (4) and the transpose identities for imaginary generators:

$$\begin{aligned} G_{TL} &= L_x^T L_x + (L_y C)^T (L_y C) = (-L_x)(L_x) + C^T L_y^T L_y C \\ &= -L_x^2 + C(-L_y)(L_y)C = \|x\|^2 I + C\|y\|^2 I\, C = \|p\|^2 I. \end{aligned}$$

Here we used $L_x^2 = -\|x\|^2 I$ and $C^2 = I$. The same computation for the bottom-right block gives $G_{BR} = \|p\|^2 I$.

Step 3 (Off-Diagonal Block F). The top-right block of G is

$$G_{TR} = L_x^T(-R_y C) + (L_y C)^T R_x = (-L_x)(-R_y C) + C^T(-L_y)R_x = L_x R_y C - C L_y R_x \;=:\; F.$$

The bottom-left block is $G_{BL} = F^T$. Therefore

$$G \;=\; \|p\|^2 I_{16} + W, \qquad W \;=\; \begin{pmatrix} 0 & F \\ F^T & 0 \end{pmatrix}. \tag{5}$$

Step 4 (The Associator Identity for F). Direct expansion: for any $z \in \mathbb{O}$,

$$F(z) = L_x R_y C(z) - C L_y R_x(z) = x(\bar{z}\, y) - \overline{y(zx)}.$$

The second term simplifies: $\overline{y(zx)} = \overline{zx} \cdot \bar{y} = (\bar{x}\bar{z})\bar{y}$. For imaginary x, y, $\bar{x} = -x$ and $\bar{y} = -y$, hence $\overline{y(zx)} = ((-x)\bar{z})(-y) = (x\bar{z})y$. Therefore

$$F(z) \;=\; x(\bar{z}\, y) \;-\; (x\bar{z})\, y \;=\; [x,\, \bar{z},\, y], \tag{6}$$

the octonion associator with middle argument $\bar{z}$.

Step 5 (Vanishing on Q: Artin's Theorem). Let $Q := \mathrm{span}\{1,\, x,\, y,\, xy\}$. We first address a degeneracy. If x, y are linearly dependent ($y = cx$ for some $c \in \mathbb{R}$), then $p = x \cdot (1, c)$ lies in an associative $\mathbb{C}$-like subalgebra of $\mathbb{S}$ in which L_p is invertible. Both sides of (3) fail trivially.

Assume henceforth x, y linearly independent. Then $\dim Q = 4$. By *Artin's theorem*, the subalgebra of $\mathbb{O}$ generated by any two elements is associative; Q is such a subalgebra. For $z \in Q$, $\bar{z} \in Q$, and the associator $[x, \bar{z}, y] = 0$ since Q is associative. By (6),

$$F\big|_Q \;\equiv\; 0, \qquad \dim(\ker F|_Q) \geq 4. \tag{7}$$

Step 6 (Associator Norm on $Q^\perp$). Let $Q^\perp \subset \mathbb{O}$ denote the four-dimensional orthogonal complement of Q. Since Q is closed under conjugation, so is $Q^\perp$; for imaginary $z \in Q^\perp$, $\bar{z} = -z \in Q^\perp$, and z satisfies $\langle z, 1\rangle = \langle z, x\rangle = \langle z, y\rangle = \langle z, xy\rangle = 0$.

Lemma 3.2 (Associator Norm on $Q^\perp$)**.** *For every $z \in Q^\perp$,*

$$\big\|[x,\, z,\, y]\big\|^2 \;=\; 4\,\|z\|^2\,(\|x\|^2\|y\|^2 - \langle x, y\rangle^2). \tag{8}$$

Proof. On $\mathrm{Im}(\mathbb{O}) \cong \mathbb{R}^7$ define the cross product $u \times v := \frac{1}{2}(uv - vu) = \mathrm{Im}(uv)$ for $u, v \in \mathrm{Im}(\mathbb{O})$, so that $uv = -\langle u, v\rangle + u \times v$. Lagrange's identity:

$$\|u \times v\|^2 \;=\; \|u\|^2\|v\|^2 - \langle u, v\rangle^2. \tag{9}$$

A direct calculation using alternativity (Baez, 2002; Schafer, 1966) gives the associator-triple-product identity

$$[x,\, z,\, y] \;=\; 2\Big(z \times (x \times y) - \langle z, x\rangle\, y + \langle z, y\rangle\, x\Big) \tag{10}$$

for $x, y, z \in \mathrm{Im}(\mathbb{O})$. Since $z \in Q^\perp$ implies $z \perp x$ and $z \perp y$, the last two terms vanish, giving $[x, z, y] = 2\, z \times (x \times y)$. Apply (9) with $u = z$, $v = x \times y$:

$$\|z \times (x \times y)\|^2 \;=\; \|z\|^2 \|x \times y\|^2 - \langle z,\, x \times y\rangle^2.$$

The second term vanishes: $x \times y = \mathrm{Im}(xy) = xy + \langle x, y\rangle \cdot 1$, so $\langle z, x \times y\rangle = \langle z, xy\rangle + \langle x, y\rangle\langle z, 1\rangle$, and both summands are zero ($z \perp xy$, $z \perp 1$). Therefore $\|z \times (x \times y)\|^2 = \|z\|^2(\|x\|^2\|y\|^2 - \langle x, y\rangle^2)$ by (9) again, and $\|[x, z, y]\|^2 = 4$ times this. □

The middle-argument $\bar{z}$ in (6) is $-z$ for imaginary z, and $Q^\perp$ is closed under conjugation. Hence $\|F(z)\|^2 = \|[x, z, y]\|^2$ for $z \in Q^\perp$:

$$\|F(z)\|^2 \;=\; 4\,\|z\|^2\,(\|x\|^2\|y\|^2 - \langle x, y\rangle^2), \qquad z \in Q^\perp. \tag{11}$$

Step 7 (Singular Values of F). Combining (7) and (11): F vanishes on the four-dimensional subspace Q, and acts on $Q^\perp$ by a fixed scalar multiple of an isometry. The squared singular values of F on $Q^\perp$ are all equal to $4(\|x\|^2\|y\|^2 - \langle x, y\rangle^2)$. Writing $\sigma := 2\sqrt{\|x\|^2\|y\|^2 - \langle x, y\rangle^2} = 2\,\|x \wedge y\|$:

$$\mathrm{SV}(F) \;=\; \{0^4,\ \sigma^4\}. \tag{12}$$

Note $\sigma = 0 \iff x, y$ linearly dependent, already handled in Step 5.

Step 8 (Spectrum of W). For an anti-diagonal block matrix $W = \begin{pmatrix} 0 & F \\ F^T & 0 \end{pmatrix}$, the eigenvalues of W are $\pm\,\mathrm{SV}(F)$. Therefore

$$\mathrm{Spec}(W) \;=\; \{-\sigma^4,\ 0^8,\ \sigma^4\}. \tag{13}$$

Step 9 (Spectrum of G). From (5): $G = \|p\|^2 I + W$, so

$$\mathrm{Spec}(G) \;=\; \{(\|p\|^2 - \sigma)^4,\ (\|p\|^2)^8,\ (\|p\|^2 + \sigma)^4\}. \tag{14}$$

Step 10 (The Keystone Identity). Since L_p has full rank iff $G = L_p^T L_p$ has trivial kernel,

$$\ker L_p \neq \{0\} \iff 0 \in \mathrm{Spec}(G) \iff \|p\|^2 = \sigma.$$

Squaring and rearranging:

$$\boxed{(\|x\|^2 - \|y\|^2)^2 \;+\; 4\langle x, y\rangle^2 \;=\; 0} \tag{15}$$

Both summands are non-negative; their sum is zero iff each vanishes individually: $\|x\| = \|y\|$ and $\langle x, y\rangle = 0$.

Step 11 (Multiplicity). When the aphanic conditions hold, (14) has smallest eigenvalue 0 with algebraic (hence geometric, G symmetric positive semidefinite) multiplicity exactly 4. Therefore $\dim\ker G = 4$. Since $G = L_p^T L_p$ acts on a finite-dimensional inner product space, $\ker L_p = \ker G$, giving $\dim\ker L_p = 4$. □

3.3 Summary

The chain of implications:

$$\underbrace{(4)}_{\text{CD}} \Rightarrow \underbrace{(5)}_{\text{composition}} \Rightarrow \underbrace{(6)}_{\text{expansion}} \overset{\text{Artin}}{\Longrightarrow} \underbrace{(7)}_{F|_Q=0} \overset{\text{Lemma 3.2}}{\Longrightarrow} \underbrace{(12)}_{\text{SV}(F)} \Rightarrow \underbrace{(14)}_{\text{Spec}(G)} \Rightarrow \underbrace{(15)}_{\text{keystone}} \Rightarrow \text{Theorem 3.1.}$$

Every link is algebraic. The composition law, skew-adjointness, and alternativity are the three pillars of $\mathbb{O}$ used at the dim-8 level. Artin's theorem handles the Q-kernel. The associator norm formula handles the $Q^\perp$ singular values. The keystone $(\|x\|^2 - \|y\|^2)^2 + 4\langle x, y\rangle^2 = 0$ forces both aphanic conditions from a single equation.

Corollary 3.3 (Determinant Formula)**.** $\det(L_p)^2 = \det(G) = (\|p\|^2 - \sigma)^4(\|p\|^2)^8(\|p\|^2 + \sigma)^4$, *with* $\sigma = 2\|x \wedge y\|$. *In particular* $\det(L_p) = 0$ *exactly on the aphanic locus, and for* $\|x\| = \|y\| = 1$, $\langle x, y\rangle = 0$ *the non-singular part gives leading coefficient* $2^4 \cdot 4^2 = 256 = 4^4$.

Remark 3.4 (On the Higher Cayley–Dickson Levels). The proof above relies on three octonion-level identities: the composition law $L_a L_{\bar{a}} = \|a\|^2 I$, alternativity (Artin's theorem in particular), and skew-adjointness on imaginaries. Of these, the composition law fails at the next CD level $\mathbb{P} = \mathbb{S} \oplus \mathbb{S} \cdot \ell'$ (this is precisely why $\mathbb{S}$ has zero divisors); alternativity fails in $\mathbb{S}$ already. What survives at every level is *flexibility* $a(ba) = (ab)a$ and the inner-product duality $\langle ab, c\rangle = \langle b, \bar{a}c\rangle$. The recursion theorem of Section 9 extends the R_ℓ-centrality result to every CD level using only these preserved identities.

4 The 4-8-4 Spectral Theorem

The proof of Theorem 3.1 produces, as intermediate output, a complete description of the spectrum of the Gram operator $G_p = L_p^\top L_p$ in the imaginary locus. We promote that output to a named theorem, since it is the structural workhorse of the holonomy and centrality results that follow.

Theorem 4.1 (4-8-4 Spectral Theorem)**.** *Let* $p = (x, y) \in \mathbb{S}$ *with* $x, y \in \mathrm{Im}(\mathbb{O})$ *linearly independent, and let* $\sigma := 2\|x \wedge y\|$. *Then the Gram operator* $G_p = L_p^\top L_p$ *has spectrum (with multiplicities)*

$$\mathrm{Spec}(G_p) \;=\; \{(\|p\|^2 - \sigma)^4,\ (\|p\|^2)^8,\ (\|p\|^2 + \sigma)^4\}. \tag{16}$$

For aphanic p *(*$\|x\| = \|y\|$ *and* $\langle x, y\rangle = 0$*), the spectrum reduces to*

$$\mathrm{Spec}(G_p) \;=\; \{0^4,\ (\|p\|^2)^8,\ (2\|p\|^2)^4\}. \tag{17}$$

For unit aphanic observers ($\|x\| = \|y\| = 1$*, hence* $\|p\|^2 = 2$*), the spectrum is the iconic* $\{0^4, 2^8, 4^4\}$.

Proof. This is equation (14) from the proof of Theorem 3.1, extracted as a stand-alone result. The aphanic-locus form follows by substituting $\|x\| = \|y\|$ and $\langle x, y\rangle = 0$ into $\sigma = 2\sqrt{\|x\|^2\|y\|^2 - \langle x, y\rangle^2}$, giving $\sigma = 2\|x\|\|y\| = \|p\|^2$. □

Remark 4.2 (Structural reading of the multiplicities). The multiplicities $4 + 8 + 4 = 16$ partition $\mathbb{S}$ as $\dim \mathbb{S}$. The 8 in the middle is the diagonal $\|p\|^2 \cdot I$ part of the CD-block decomposition (Step 2 of the characterization proof). The two 4's are the symmetric pair $\pm\sigma$ contributed by the off-diagonal block W in (5); they originate from the eight singular values of W being arranged as $\pm\,\mathrm{SV}(F)$ with $\mathrm{SV}(F) = \{0^4, \sigma^4\}$. Equivalently, the 4's come from the four-dimensional associative subalgebra $Q = \mathrm{span}\{1, x, y, xy\}$ and its rank-4 complement $Q^\perp$ inside $\mathbb{O}$.

Corollary 4.3 (Determinant and Pfaffian)**.** $\det(L_p)^2 = (\|p\|^2)^8(\|p\|^2-\sigma)^4(\|p\|^2+\sigma)^4 = (\|p\|^2)^8((\|p\|^2)^2-\sigma^2)^4$. *In particular* $\det(L_p)=0$ *exactly on the aphanic locus. Normalizing* $\|x\|=\|y\|=1$ *(so that* $\|p\|^2=2$ *and* $\sigma\to 2$*) and writing* $\varepsilon := \|p\|^2-\sigma$ *for the approach parameter, the displayed identity gives* $\det(L_p)^2 = 2^8\,\varepsilon^4\,4^4 + O(\varepsilon^5)$*, so the leading non-vanishing coefficient of* $\det(L_p)$ *is* $\sqrt{2^8\cdot 4^4} = 2^4\cdot 4^2 = 4^4 = 256$*. The sign is determined, not a choice:* L_p *is real skew-symmetric of even order for imaginary* p*, so* $\det L_p = \mathrm{Pf}(L_p)^2 \geq 0$ *(Lemma 5.3, proved in Section 5 below), and the positive root is the only admissible one. The normalization is needed: without it the coefficient is* $(\|p\|^2)^4(\|p\|^2+\sigma)^2$.

Corollary 4.4 (Spectral projectors)**.** *For aphanic* p *the three eigenvalues* $\{0,\|p\|^2,2\|p\|^2\}$ *are distinct, so each spectral projector of* G_p *is a polynomial in* G_p *given by Lagrange interpolation. In particular the kernel projector* π_p *onto* $\ker L_p$ *is*

$$\pi_p \;=\; \frac{(G_p-\|p\|^2 I)(G_p-2\|p\|^2 I)}{(0-\|p\|^2)(0-2\|p\|^2)} \;=\; \frac{1}{2\|p\|^4}\,(G_p-\|p\|^2 I)(G_p-2\|p\|^2 I).$$

Any operator commuting with G_p *commutes with every spectral projector, hence in particular with* π_p.

The spectral projector formula above is what allows Theorem 8.1 (R_ℓ centrality on G_p) to descend to a centrality identity on π_p, recovering the original fiberwise centrality statement as a corollary.

Remark 4.5 (Status). PROVEN. Equivalent to (14) from Section 3.

5 The Sedenion Annihilator Fourfold Theorem

The Characterization Theorem 3.1 and the spectral result of Section 4 were stated for a doubly imaginary generator $p=(x,y)$ with $x,y\in\mathrm{Im}(\mathbb{O})$. Two natural questions are left open by that hypothesis. First, what happens when the second component is permitted a real part, i.e. $p=(a,b)$ with $a\in\mathrm{Im}(\mathbb{O})$ but $b\in\mathbb{O}$ arbitrary? Second, what is the left-annihilator behaviour of a *completely* general nonzero sedenion, including those with nonzero real part? This section answers both. The outcome is a single clean dichotomy for every nonzero $p\in\mathbb{S}$, together with the exact left-nullity spectrum and the equality of left and right nullities. We refer to the combined statement as the *Annihilator Fourfold Theorem* (Theorem 5.14).

Relation to prior work. The zero divisors of the sedenions have been studied previously from a set-theoretic and structural standpoint. In proposing their programme, Biss et al. (2008, §1) observed that the zero-divisor loci of the Cayley–Dickson algebras should prove geometrically interesting objects in their own right, and set the understanding of the topological properties of $\mathrm{ZD}(\mathcal{A}_n)$ as an aim. The present paper carries that programme out for the $\mathcal{A}_4$ stratum, by studying not the locus alone but the rank-4 kernel bundle it carries. Moreno (1998) determined the zero-divisor locus of the Cayley–Dickson algebras over $\mathbb{R}$, showing in particular that a doubly imaginary $p=(x,y)$ is a zero divisor exactly when $\|x\|=\|y\|$ and $\langle x,y\rangle=0$ (the two-condition criterion, re-proved by elementary means as Theorem 3.1), and described the resulting variety as stratified by the rank of L_p. Cawagas (2004) analysed the subalgebra structure of $\mathbb{S}$ and the organisation of its zero divisors.

The structural facts recorded in this section are, individually, known. Moreno (1998, Cor. 1.17) proved that $\dim_{\mathbb{R}}\ker L_p$ is always a multiple of 4, and gave the zero-divisor criterion (Moreno, 1998, Cor. 2.14); the criterion was obtained independently by Khalil and Yiu (1997, Thm. 3.2.3). Biss et al. (2008, Thm. 1.2) proved the bound $\dim\mathrm{Ann}(x)\leq 2^n-4n+4$, which at $n=4$ equals 4, re-proved Moreno's divisibility (Biss et al., 2008, Thm. 9.8), and stated the resulting $\{0,4\}$ dichotomy for $\mathbb{S}$ explicitly (Biss et al., 2008, §12), where the criterion appears as (Biss et al., 2008, Prop. 12.1). In particular the imaginarity of *both* components — and with it

$\Re(b) = 0$ — is a *consequence* of the zero-divisor hypothesis in that work (Biss et al., 2008, Lem. 9.4), not an assumption; we claim no improvement on that point. Equality of the left and right annihilators holds in every Cayley–Dickson algebra (Biss et al., 2008, Cor. 9.3), which subsumes $\dim_{\mathbb{R}} \ker L_p = \dim_{\mathbb{R}} \ker R_p$ for $\mathbb{S}$. The determinant factorization below was obtained independently, in the equivalent form $\det L_v = D_1^4 D_2^2$, by Koebisu (2026, Thm. 3.7).

What this section contributes is therefore not the dichotomy but the *route*: a single Pfaffian identity, $\mathrm{Pf}(L_p) = -(\|p\|^2)^2 T$, certified by an exact finite computation, from which the criterion, the determinant factorization $\det L_p = (\|p\|^2)^4 T^2$, and the nullity value all follow in one derivation. That route is what carries into the spectral and bundle-theoretic material of Sections 4 and 6, which is independent of the algebraic literature cited here.

Throughout, $L_p, R_p \in \mathrm{End}(\mathbb{S})$ are left and right multiplication in the standard ordered basis $\{1, e_1, \ldots, e_{15}\}$, $\ell = e_8$ is the doubling generator, and $p = (a, b)$ abbreviates $p = \iota_1(a) + \iota_2(b)$ with $a, b \in \mathbb{O}$. We write $|\cdot|$ for the octonion norm, $\langle\,,\rangle$ for its inner product, and $\Re(b) = b_0$ for the real part. We use repeatedly the three octonion-level facts collected in Section 3.1: the composition law $L_u L_{\bar{u}} = R_u R_{\bar{u}} = |u|^2 I$, skew-adjointness of imaginary generators ($L_w^\top = -L_w$ for $w \in \mathrm{Im}(\mathbb{O})$), and alternativity of $\mathbb{O}$. At the sedenion level we use only the two tower-universal properties of Table 1: flexibility and the inner-product duality $\langle uv, w\rangle = \langle v, \bar{u}w\rangle$, which gives $L_p^\top = L_{\bar{p}}$ on $\mathbb{S}$.

5.1 Elementary reductions

Lemma 5.1 (Real part forces trivial kernel)**.** *If $p \in \mathbb{S}$ has $\Re(p) \neq 0$, then L_p is invertible; in particular* $\ker L_p = \{0\}$.

Proof. Write $p = p_0 + p_{\mathrm{im}}$ with $p_0 = \Re(p) \in \mathbb{R}$ and $p_{\mathrm{im}} \in \mathrm{Im}(\mathbb{S})$, so $L_p = p_0 I + L_{p_{\mathrm{im}}}$. The inner-product duality $\langle uv, w\rangle = \langle v, \bar{u}w\rangle$ holds at every Cayley–Dickson level and gives $L_q^\top = L_{\bar{q}}$ for all $q \in \mathbb{S}$; for $q = p_{\mathrm{im}} \in \mathrm{Im}(\mathbb{S})$ we have $\bar{q} = -q$, hence $L_{p_{\mathrm{im}}}^\top = -L_{p_{\mathrm{im}}}$. Thus $L_{p_{\mathrm{im}}}$ is a real skew-symmetric matrix, and its eigenvalues are purely imaginary. The eigenvalues of $L_p = p_0 I + L_{p_{\mathrm{im}}}$ are therefore of the form $p_0 + i\mu$ with $\mu \in \mathbb{R}$, none of which vanishes when $p_0 \neq 0$. Hence $\det L_p \neq 0$. □

By Lemma 5.1 the annihilator problem is nontrivial only for imaginary p, i.e. $\Re(p) = 0$. In our coordinates $\Re(p) = \Re(a)$, so p *imaginary means* $a \in \mathrm{Im}(\mathbb{O})$ *while* $b \in \mathbb{O}$ *is unconstrained.* This is the precise sense in which the present section generalizes Section 3: there b was assumed imaginary, here it is arbitrary. We record the two genuinely degenerate cases.

Lemma 5.2 (Degenerate components)**.** *Let $p = (a, b)$ be imaginary and nonzero.*

(i) *If $a = 0$ then* $\ker L_p = \{0\}$.

(ii) *If $b = 0$ then* $\ker L_p = \{0\}$.

Consequently $\ker L_p \neq \{0\}$ *forces $a \neq 0$ and $b \neq 0$.*

Proof. If $a = 0$ then $p = (0, b)$ with $b \neq 0$, and the product $(0, b)(u, v) = (-\bar{v}b,\, b\bar{u})$ vanishes only if $\bar{v}b = 0$ and $b\bar{u} = 0$; since $\mathbb{O}$ is a division algebra and $b \neq 0$, this forces $v = u = 0$. If $b = 0$ then $p = (a, 0)$ with $a \in \mathrm{Im}(\mathbb{O}) \setminus \{0\}$, and $(a, 0)(u, v) = (au,\, va)$ vanishes only if $au = va = 0$, again forcing $u = v = 0$ in the division algebra $\mathbb{O}$. Both kernels are trivial. □

5.2 The determinant and Pfaffian identity

For imaginary p the operator L_p is skew-symmetric, so its determinant is a perfect square and is controlled by the Pfaffian.

Lemma 5.3 (Skew-symmetry and the Pfaffian)**.** *For imaginary $p \in \mathbb{S}$, $L_p^\top = -L_p$, so L_p is a 16×16 real skew-symmetric matrix and*

$$\det L_p \;=\; \mathrm{Pf}(L_p)^2, \tag{18}$$

where Pf *is the Pfaffian taken in the standard ordered basis.*

Proof. $L_p^\top = L_{\bar{p}} = L_{-p} = -L_p$ as in Lemma 5.1. Identity (18) is the defining property of the Pfaffian of a skew-symmetric matrix of even order. □

Introduce the four real invariants of $p = (a, b)$,

$$I_1 = |a|^2, \quad D_1 = \Re(b), \quad D_2 = \langle a, b\rangle, \quad D_3 = |a|^2 - |b|^2, \quad |p|^2 = |a|^2 + |b|^2, \tag{19}$$

and the nonnegative quantity

$$T \;=\; 4\,|a|^2\Re(b)^2 \;+\; 4\,\langle a, b\rangle^2 \;+\; \left(|a|^2 - |b|^2\right)^2 \;=\; 4I_1D_1^2 + 4D_2^2 + D_3^2. \tag{20}$$

Proposition 5.4 (Determinant/Pfaffian factorization)**.** *For every imaginary $p = (a, b) \in \mathbb{S}$ (that is, $a \in \mathrm{Im}(\mathbb{O})$, $b \in \mathbb{O}$),*

$$\mathrm{Pf}(L_p) \;=\; -\,(|p|^2)^2\,T, \qquad \textit{hence} \qquad \det L_p \;=\; (|p|^2)^4\,T^2, \tag{21}$$

with T as in (20). Both sides are polynomials in the sixteen real coordinates of p; the displayed sign of Pf *is that of the standard ordered basis, while* $\det L_p$ *is independent of orientation.*

The proof is given in Section 5.8 as an *exact finite certificate of a polynomial identity*: both sides are G_2-invariant polynomials, the Pfaffian has total degree 8, and an exact evaluation on a complete 9-point-per-variable grid in normal-form coordinates determines the identity over $\mathbb{Z}$. The certificate is purely algebraic (exact integer and modular arithmetic, no floating-point inference). We first record that the identity is consistent with, and extends, the spectral results already proved.

Remark 5.5 (Consistency with the 4-8-4 spectrum). When $b \in \mathrm{Im}(\mathbb{O})$ one has $\Re(b) = 0$, so $D_1 = 0$ and $T = (|a|^2 - |b|^2)^2 + 4\langle a, b\rangle^2$. A short computation gives $T = (|p|^2)^2 - \sigma^2$ with $\sigma = 2\sqrt{|a|^2|b|^2 - \langle a, b\rangle^2}$ the wedge norm of Section 4:

$$(|p|^2)^2 - \sigma^2 = (|a|^2 + |b|^2)^2 - 4(|a|^2|b|^2 - \langle a, b\rangle^2) = (|a|^2 - |b|^2)^2 + 4\langle a, b\rangle^2.$$

Hence (21) reduces to $\det L_p = (|p|^2)^4\big((|p|^2)^2 - \sigma^2\big)^2$, equivalently $\det(L_p)^2 = (|p|^2)^8\big((|p|^2)^2 - \sigma^2\big)^4$, which is exactly the Determinant Corollary 3.3 obtained from the singular values $\mathrm{SV}(F) = \{0^4, \sigma^4\}$ in Step 7 of the proof of Theorem 3.1. The new content of Proposition 5.4 is the additional term $4|a|^2\Re(b)^2$, which records the effect of a nonzero real part of b and is invisible to the imaginary-locus analysis.

5.3 The annihilator criterion

Theorem 5.6 (Annihilator criterion)**.** *Let $p = (a, b) \in \mathbb{S}$ be imaginary and nonzero. Then*

$$\ker L_p \neq \{0\} \iff \Re(b) = 0, \quad \langle a, b\rangle = 0, \quad |a| = |b|. \tag{22}$$

Proof. $\ker L_p \neq \{0\}$ iff $\det L_p = 0$. By Proposition 5.4, $\det L_p = (|p|^2)^4T^2$, so $\det L_p = 0$ iff $|p|^2 = 0$ or $T = 0$. As $p \neq 0$ we have $|p|^2 > 0$, whence $\det L_p = 0 \iff T = 0$.

The expression $T = 4|a|^2\Re(b)^2 + 4\langle a, b\rangle^2 + (|a|^2 - |b|^2)^2$ is a sum of three nonnegative real numbers, so $T = 0$ holds iff each summand vanishes:

$$|a|^2\,\Re(b)^2 = 0, \qquad \langle a, b\rangle = 0, \qquad |a|^2 = |b|^2.$$

By Lemma 5.2, $\ker L_p \neq \{0\}$ forces $a \neq 0$, hence $|a|^2 > 0$; the first equation then gives $\Re(b) = 0$. This proves the forward implication. Conversely, if $\Re(b) = 0$, $\langle a, b\rangle = 0$ and $|a| = |b|$, then every summand of T vanishes, so $T = 0$, $\det L_p = 0$, and $\ker L_p \neq \{0\}$. □

Theorem 5.6 recovers the equivalence (3) of Theorem 3.1 without assuming b imaginary, deriving $\Re(b) = 0$ from the vanishing of T; the same implication is established by a direct argument in (Biss et al., 2008, Lem. 9.4). What is particular to the present derivation is its source: the non-negativity of the three summands of the single invariant T, which is the analogue, for the full invariant, of the keystone identity (15).

5.4 The kernel dimension is exactly four

When the criterion of Theorem 5.6 holds, the second component b is imaginary, $\langle a, b\rangle = 0$, $|a| = |b|$, and (by Lemma 5.2) a, b are nonzero, hence linearly independent. Thus (a, b) is an aphanic pair in the sense of Section 3, and the dimension follows immediately from the algebraic characterization.

Theorem 5.7 (Exact kernel dimension)**.** *If $p = (a, b)$ is imaginary with $\ker L_p \neq \{0\}$, then $\dim_{\mathbb{R}} \ker L_p = 4$.*

Proof. By Theorem 5.6, $\Re(b) = 0$, so $b \in \mathrm{Im}(\mathbb{O})$, and $|a| = |b|$, $\langle a, b\rangle = 0$ with a, b linearly independent. These are exactly the hypotheses of Theorem 3.1 (aphanic case), which gives $\dim_{\mathbb{R}} \ker L_p = 4$. □

The dimension claim therefore needs nothing beyond the octonion-level algebra of Section 3. For completeness, and because it exposes the G_2-symmetry on which Section 6 rests, we record an independent route to the same value: a quaternionic action on the kernel forces $4 \mid \dim_{\mathbb{R}} \ker L_p$, and a single exact rank computation pins the value to 4.

Lemma 5.8 (Quaternionic operators on the kernel)**.** *Let $p = (a, b)$ be imaginary with $|a| = |b|$ and $\langle a, b\rangle = 0$, and define $J_1 = R_{(a,0)}$ and $J_2 = R_{(b,0)}$ on $\mathbb{S}$, i.e. right multiplication by the lower-embedded halves. Explicitly $J_1(u, v) = (ua, v\bar{a})$ and $J_2(u, v) = (ub, v\bar{b})$. Then, as operators on $\mathbb{S}$,*

$$J_1^2 = J_2^2 = -\,|a|^2\, I, \qquad J_1 J_2 + J_2 J_1 = 0. \tag{23}$$

Proof. Both identities are computed entirely within $\mathbb{O}$, where multiplication is alternative, so every step below is a legitimate octonion identity. For the squares, the right-alternative law $(wc)c = wc^2$ gives $J_1^2(u, v) = ((ua)a, (v\bar{a})\bar{a}) = (ua^2, v\bar{a}^2)$; since $a \in \mathrm{Im}(\mathbb{O})$, $a^2 = -|a|^2$ and $\bar{a}^2 = (-a)^2 = -|a|^2$, so $J_1^2 = -|a|^2 I$. The same computation for b gives $J_2^2 = -|b|^2 I$, and $|b| = |a|$ yields $J_2^2 = -|a|^2 I$.

For the anticommutator, consider the first component of $(J_1 J_2 + J_2 J_1)(u, v)$:

$$(ub)a + (ua)b = u(ba + ab) + \big([u, b, a] + [u, a, b]\big),$$

where $[u, c, c'] = (uc)c' - u(cc')$ is the octonion associator. In the alternative algebra $\mathbb{O}$ the associator is alternating in all three arguments over the whole of $\mathbb{O}$ (not only on $\mathrm{Im}\,\mathbb{O}$), so $[u, b, a] + [u, a, b] = 0$; and $ab + ba = -2\langle a, b\rangle = 0$ for orthogonal imaginary a, b. Hence the first component vanishes. The second component, $(v\bar{b})\bar{a} + (v\bar{a})\bar{b}$, vanishes by the identical computation with $\bar{a} = -a$, $\bar{b} = -b$. Thus $J_1 J_2 + J_2 J_1 = 0$. □

The relations (23) hold globally on $\mathbb{S}$. To make $\ker L_p$ an $\mathbb{H}$-module it remains to show J_1, J_2 *preserve* the kernel; this is established in Proposition 5.11 below, by an explicit check at the canonical element followed by G_2-transport.

5.5 G_2-transport and the canonical kernel

The automorphism group $G_2 = \mathrm{Aut}(\mathbb{O})$ acts on $\mathbb{S}$ diagonally (Lemma 6.2): for $\varphi \in G_2$, the map $\widetilde{\varphi} = \mathrm{diag}(\varphi, \varphi)$ is an algebra automorphism of $\mathbb{S}$, whence

$$L_{\widetilde{\varphi}p} = \widetilde{\varphi}\, L_p\, \widetilde{\varphi}^{-1}, \qquad \ker L_{\widetilde{\varphi}p} = \widetilde{\varphi}(\ker L_p), \qquad \widetilde{\varphi}(a, 0) = (\varphi a, 0). \tag{24}$$

In particular $\dim_{\mathbb{R}} \ker L_p$ is constant on G_2-orbits, and J_1, J_2 are carried to the corresponding operators at $\widetilde{\varphi} p$.

Proposition 5.9 (Reduction to the canonical pair)**.** *Every imaginary $p = (a, b)$ with $|a| = |b| \neq 0$, $\langle a, b\rangle = 0$, and $b \in \mathrm{Im}(\mathbb{O})$ is carried by a scaling and a diagonal G_2-automorphism to the canonical element $p_0 = e_1 + e_{10} = (e_1, e_2)$. Consequently J_1, J_2 preserve $\ker L_p$ for all such p if and only if they preserve $\ker L_{p_0}$.*

Proof. Scaling $p \mapsto p/|a|$ multiplies L_p by the positive scalar $|a|$, hence fixes $\ker L_p$ and rescales J_1, J_2; it reduces to the case $|a| = |b| = 1$, i.e. (a, b) an ordered pair of orthonormal imaginary octonions. By Baez (2002, §4.1), G_2 acts *simply transitively on basic triples*: for any two ordered triples (f_1, f_2, f_3) of imaginary octonions with $f_i^2 = -1$, $\langle f_1\rangle \cong \mathbb{C}$, $\langle f_1, f_2\rangle \cong \mathbb{H}$, and f_1, f_2, f_3 generating $\mathbb{O}$, there is a unique $\varphi \in G_2$ carrying the first to the second. An ordered orthonormal imaginary pair (a, b) satisfies $a^2 = b^2 = -1$ and $a \perp b$, so $\langle a, b\rangle \cong \mathbb{H}$; choosing any unit c orthogonal to the four-dimensional subalgebra $\langle 1, a, b, ab\rangle$ (its orthogonal complement in $\mathrm{Im}(\mathbb{O})$ has dimension 4, hence is nonempty) makes (a, b, c) a basic triple. Simple transitivity on basic triples therefore specializes to transitivity on ordered orthonormal imaginary 2-frames, so $(a, b) = (\varphi e_1, \varphi e_2)$ for some $\varphi \in G_2$, and $\widetilde{\varphi} p_0 = p$ after scaling. The equivalence of the invariance statements is then immediate from (24). □

Lemma 5.10 (Exact canonical rank)**.** *At the canonical element $p_0 = e_1 + e_{10}$ one has $\operatorname{rank} L_{p_0} = 12$ and $\dim_{\mathbb{R}} \ker L_{p_0} = 4$. An explicit orthogonal basis of $\ker L_{p_0}$ is*

$$e_7 + e_{12}, \quad -e_6 + e_{13}, \quad e_5 + e_{14}, \quad -e_4 + e_{15}, \tag{25}$$

equivalently, in half-coordinates, (e_7, e_4), $(-e_6, e_5)$, (e_5, e_6), $(-e_4, e_7)$. Writing v_1, v_2, v_3, v_4 for this basis, the operators $J_1 = R_{(e_1,0)}$ and $J_2 = R_{(e_2,0)}$ of Lemma 5.8 preserve $\ker L_{p_0}$, acting by

$$J_1 : \ v_1 \mapsto v_2 \mapsto -v_1, \quad v_3 \mapsto -v_4 \mapsto v_3; \qquad J_2 : \ v_1 \mapsto v_3 \mapsto -v_1, \quad v_2 \mapsto v_4 \mapsto -v_2, \tag{26}$$

so that $J_1^2 = J_2^2 = -I$ and $J_1 J_2 + J_2 J_1 = 0$ on $\ker L_{p_0}$.

Proof. The four vectors in (25) have pairwise disjoint support, hence are linearly independent, and a direct application of the multiplication (2) verifies $p_0 \cdot v = 0$ for each; thus $\dim_{\mathbb{R}} \ker L_{p_0} \geq 4$. For the reverse bound, the 12×12 leading minor of L_{p_0} on rows and columns $\{0, 1, \ldots, 11\}$ has determinant $16 \neq 0$ (an exact integer computation), so $\operatorname{rank} L_{p_0} \geq 12$, i.e. $\dim_{\mathbb{R}} \ker L_{p_0} \leq 4$. The two bounds give equality. The action (26) is the direct evaluation of $v \mapsto v \cdot (e_1, 0)$ and $v \mapsto v \cdot (e_2, 0)$ via (2): each image is again one of $\pm v_i$, so $\ker L_{p_0}$ is preserved, and (26) gives $J_1^2 = J_2^2 = -I$ and $\{J_1, J_2\} = 0$ on the kernel by inspection. Every entry of L_{p_0} is an integer in $\{-1, 0, 1\}$, so the determinant, the kernel relations, and the action (26) are verified by exact integer arithmetic with no rounding. □

Combining the base-point value $\dim_{\mathbb{R}} \ker L_{p_0} = 4$ (Lemma 5.10) with the reduction to p_0 (Proposition 5.9) and the orbit-invariance of $\dim_{\mathbb{R}} \ker L_p$ from (24) gives a second proof that $\dim_{\mathbb{R}} \ker L_p = 4$ for every imaginary p with nontrivial kernel, in agreement with Theorem 5.7. This route is independent of the octonion-level multiplicity count of Section 3 on which Theorem 5.7 rests, although – unlike that primary argument – it does invoke the G_2-transitivity fact of Proposition 5.9. Its value lies less in re-proving the dimension than in exposing the quaternionic structure of the kernel, recorded next, which underlies the bundle of Section 6.

Proposition 5.11 (Quaternionic module structure)**.** *For imaginary $p = (a, b)$ with $\ker L_p \neq \{0\}$, the operators $J_1 = R_{(a,0)}$ and $J_2 = R_{(b,0)}$ preserve $\ker L_p$, which is thereby a left module over the quaternion algebra generated by the identity and the normalized operators $J_1/|a|, J_2/|a|$. Consequently $4 \mid \dim_{\mathbb{R}} \ker L_p$.*

Proof. Invariance holds at the canonical element $p_0 = e_1 + e_{10}$ by the explicit action (26) of Lemma 5.10: J_1, J_2 map the kernel basis into itself. By Theorem 5.6 and Proposition 5.9, any p with $\ker L_p \neq \{0\}$ equals, after scaling, $\widetilde{\varphi} p_0$ for some $\varphi \in G_2$. Since $\widetilde{\varphi}(a, 0) = (\varphi a, 0)$, the canonical operators intertwine, $\widetilde{\varphi}\, R_{(e_1,0)}\, \widetilde{\varphi}^{-1} = R_{(\varphi e_1,0)}$ and likewise for e_2; together with $\ker L_p = \widetilde{\varphi}(\ker L_{p_0})$ from (24), this carries the invariance at p_0 to invariance at p. With the global relations (23) (Lemma 5.8), the normalized $J_1/|a|, J_2/|a|$ make $\ker L_p$ a finite-dimensional real $\mathbb{H}$-module; such a module is free over the division ring $\mathbb{H}$, so $4 \mid \dim_{\mathbb{R}} \ker L_p$. □

5.6 The nullity spectrum and left–right equality

Theorem 5.12 (Nullity spectrum)**.** *For nonzero $p \in \mathbb{S}$ the possible values of $\dim_{\mathbb{R}} \ker L_p$ are exactly 0 and 4. Both occur: $\dim_{\mathbb{R}} \ker L_p = 0$ whenever $\Re(p) \neq 0$ (Lemma 5.1) and, for imaginary p, whenever the criterion of Theorem 5.6 fails; $\dim_{\mathbb{R}} \ker L_p = 4$ precisely on the imaginary aphanic locus. In particular $4 \mid \dim_{\mathbb{R}} \ker L_p$ for every nonzero p.*

Proof. If $\Re(p) \neq 0$ then $\ker L_p = \{0\}$ by Lemma 5.1. If p is imaginary, then either the criterion of Theorem 5.6 fails and $\ker L_p = \{0\}$, or it holds and $\dim_{\mathbb{R}} \ker L_p = 4$ by Theorem 5.7. No other value occurs. The value 0 is realized by, e.g., any p with $\Re(p) \neq 0$, and the value 4 by the canonical $p_0 = e_1 + e_{10}$ (Lemma 5.10). □

Theorem 5.13 (Left–right equality)**.** *For every $p \in \mathbb{S}$, $\dim_{\mathbb{R}} \ker L_p = \dim_{\mathbb{R}} \ker R_p$.*

Proof. Conjugation $x \mapsto \bar{x}$ is a real-linear isometric involution of $\mathbb{S}$ and an algebra anti-homomorphism, $\overline{xy} = \bar{y}\,\bar{x}$ at every Cayley–Dickson level. Hence $px = 0 \iff \overline{px} = \bar{x}\,\bar{p} = 0$, i.e. $x \in \ker L_p \iff \bar{x} \in \ker R_{\bar{p}}$. The map $x \mapsto \bar{x}$ is a real-linear bijection $\ker L_p \to \ker R_{\bar{p}}$, so $\dim_{\mathbb{R}} \ker L_p = \dim_{\mathbb{R}} \ker R_{\bar{p}}$. For imaginary p, $\bar{p} = -p$ and $R_{\bar{p}} = -R_p$, which has the same kernel as R_p; for p with $\Re(p) \neq 0$ both kernels are trivial by the analogue of Lemma 5.1 for R_p (right multiplication by an imaginary element is also skew). In either case $\dim_{\mathbb{R}} \ker L_p = \dim_{\mathbb{R}} \ker R_p$. □

5.7 The main theorem

Theorem 5.14 (Sedenion Annihilator Fourfold Theorem)**.** *Let $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}$ be the sedenion algebra under the Cayley–Dickson product (2), and for $p = (a, b) \in \mathbb{S}$ let $L_p(x) = px$. For every nonzero p:*

(1) *if $\Re(p) \neq 0$, then $\ker L_p = \{0\}$;*

(2) *if p is imaginary (so $a \in \mathrm{Im}(\mathbb{O})$, $b \in \mathbb{O}$), then*

$$\ker L_p \neq \{0\} \iff \Re(b) = 0,\ \langle a, b\rangle = 0,\ |a| = |b|;$$

(3) *the real dimension of $\ker L_p$ is either 0 or 4; in particular $4 \mid \dim_{\mathbb{R}} \ker L_p$;*

(4) $\dim_{\mathbb{R}} \ker L_p = \dim_{\mathbb{R}} \ker R_p$.

Proof. Part (1) is Lemma 5.1; part (2) is Theorem 5.6; part (3) combines Theorem 5.12 with Theorem 5.7; part (4) is Theorem 5.13. □

Remark 5.15 (Status and scope). PROVEN. The dimension value 4 (and hence the spectrum $\{0, 4\}$) rests on the octonion-level Characterization Theorem 3.1 and is algebraic throughout. The determinant identity Proposition 5.4 is established by the exact finite certificate of Section 5.8; its imaginary-component restriction coincides with the spectral Corollary 3.3 (Remark 5.5). The independent G_2-transport route of Section 5.5 invokes one standard external fact, the simple transitivity of G_2 on basic triples (Baez, 2002, §4.1); the primary dimension argument does not use it. The theorem is a statement about the sedenion algebra $\mathbb{S}$ alone: no claim is made about the pathion algebra or any higher Cayley–Dickson level, where the determining octonion identities (composition, alternativity) no longer hold.

5.8 An exact certificate for the Pfaffian identity

This subsection proves Proposition 5.4. The argument is a finite *exact* certificate: it reduces the identity to a finite list of exact integer evaluations, with a proven degree bound guaranteeing that the list is complete. No floating-point arithmetic and no random sampling enter the proof.

Both sides are G_2-invariant. For $\varphi \in G_2$ the diagonal automorphism $\widetilde{\varphi}$ lies in SO(16) (it is orthogonal and fixes the real axis), and by (24) $L_{\widetilde{\varphi}p} = \widetilde{\varphi} L_p \widetilde{\varphi}^{\top}$. For a skew matrix M and $S \in \mathrm{SO}(16)$ one has $\mathrm{Pf}(SMS^{\top}) = \det(S)\,\mathrm{Pf}(M) = \mathrm{Pf}(M)$, so $\mathrm{Pf}(L_p)$ is G_2-invariant; the invariants $I_1, D_1, D_2, D_3, |p|^2$ in (19) are manifestly G_2-invariant, so the right-hand side $-(|p|^2)^2 T$ is as well.

Reduction to a normal form. Every imaginary $p = (a, b)$ with $a \neq 0$ lies in the G_2-orbit of a normal form

$$p = (\alpha e_1,\ \beta + \gamma e_1 + \delta e_2), \qquad \alpha, \beta, \gamma, \delta \in \mathbb{R}, \tag{27}$$

obtained by rotating a to the e_1-axis (transitivity of G_2 on imaginary directions) and then rotating the part of b orthogonal to the $\langle 1, e_1 \rangle$ plane into the e_2-direction (the stabilizer SU(3) of e_1 acts transitively on spheres in the orthogonal 6-space). The achievable invariants are $I_1 = \alpha^2$, $D_1 = \beta$, $D_2 = \alpha\gamma$, $D_3 = \alpha^2 - (\beta^2 + \gamma^2 + \delta^2)$; the constraint $\delta^2 = |b|^2 - \Re(b)^2 - \langle a, b\rangle^2/|a|^2 \geq 0$ is the Cauchy–Schwarz inequality, so (27) realizes every orbit (the case $a = 0$ occurs at $\alpha = 0$). Since both sides of (21) are G_2-invariant, it suffices to prove the identity on the normal forms (27), i.e. as a polynomial identity in the four variables $(\alpha, \beta, \gamma, \delta)$.

Degree bound. Each entry of L_p is linear in the coordinates of p, so on the normal form (27) each entry is linear in $(\alpha, \beta, \gamma, \delta)$. The Pfaffian of a 16×16 matrix is a homogeneous polynomial of degree 8 in the matrix entries; hence $\mathrm{Pf}(L_p)$ is a polynomial of total degree 8, and therefore of degree at most 8 in each of $\alpha, \beta, \gamma, \delta$. The right-hand side $-(|p|^2)^2 T$ is likewise of degree 8. Two polynomials of degree at most 8 in each variable that agree at 9 distinct values of each variable – that is, on the full product grid $\{-4, -3, \ldots, 4\}^4$ of $9^4 = 6561$ points – are identical (multivariate Lagrange interpolation / the Combinatorial Nullstellensatz). It therefore suffices to verify (21) at these 6561 points by exact arithmetic.

Exact evaluation of the Pfaffian. At an integer grid point the entries of L_p are integers, and on the normal form each entry is a single signed coordinate, so $|(L_p)_{ij}| \leq 4$. The Pfaffian is a signed sum of the $15!! = 2{,}027{,}025$ perfect-matching products of eight entries, whence

$$|\,\mathrm{Pf}(L_p)| \ \leq\ 15!! \cdot 4^8 \ =\ 132{,}843{,}110{,}400 \ <\ \tfrac{1}{2} P_1 P_2, \tag{28}$$

where $P_1 = 2{,}000{,}000{,}011$ and $P_2 = 1{,}000{,}000{,}007$ are two primes with $P_1 P_2 \approx 2 \times 10^{18}$. Computing $\mathrm{Pf}(L_p)$ mod P_1 and mod P_2 by exact skew-symmetric Gaussian elimination over the respective prime fields and reconstructing by the Chinese Remainder Theorem yields a unique integer in $(-\tfrac{1}{2}P_1P_2, \tfrac{1}{2}P_1P_2)$; by the bound (28) this integer *is* $\mathrm{Pf}(L_p)$. The right-hand side $-(|p|^2)^2 T$ is a small integer evaluated directly. Carrying this out at all 6561 grid points, the two integers coincide at every point. By the degree bound this proves $\mathrm{Pf}(L_p) = -(|p|^2)^2 T$ as a polynomial identity over $\mathbb{Z}$, and squaring gives $\det L_p = (|p|^2)^4 T^2$. □

Remark 5.16 (On the certificate). The computation is a finite, exact, and independently reproducible verification of an algebraic identity, not numerical evidence: the arithmetic is over $\mathbb{Z}$ and finite fields, the degree bound is proved, the evaluation grid is complete for that degree, and the integer Pfaffian is recovered exactly under the bound (28). The same value of $\mathrm{Pf}(L_p)$ is obtained from either prime alone within (28), and $\mathrm{Pf}(L_p)^2 = \det L_p$ provides a further consistency check against an independent determinant routine. A self-contained program performing the certificate is included in the supplementary archive.

6 Bundle Topology and the Aphanic Twist

By Theorem 3.1, $\ker L_p$ is a four-dimensional subspace of $\mathbb{S}$ for every aphanic p. As p varies over the aphanic locus $\mathcal{V}$, the kernels assemble into a continuous family — a rank-4 real vector bundle $\mathcal{K} \to \mathcal{V}$. This section establishes two structural facts: (1) the bundle is G_2-equivariant with quaternionic fibers (the structure group reduces to SU(2) acting via the fundamental representation), and (2) the bundle is topologically nontrivial, carrying a nonzero $\mathbb{Z}/2$ secondary characteristic class. The non-flatness of the canonical connection follows as a corollary.

6.1 The Aphanic Locus as a Homogeneous Space

Proposition 6.1 (Aphanic locus is a G_2-orbit)**.** *The aphanic locus $\mathcal{V} \subset \mathbb{S}$ (unit-normalized, per Section 2.3) is a single G_2-orbit, diffeomorphic to the homogeneous space $G_2/\mathrm{SU}(2)$. The dimension is $\dim G_2 - \dim \mathrm{SU}(2) = 14 - 3 = 11$. The identification of the unit zero-divisor locus of $\mathbb{S}$ with the Stiefel manifold $V_2(\mathbb{R}^7)$ is due to Moreno (1998) and is recorded again in (Biss et al., 2008, §12); the homogeneous-space presentation $G_2/\mathrm{SU}(2)$ used here, and everything bundle-theoretic built on it in Sections 6–9, is what the present treatment adds.*

Sketch. $G_2 = \mathrm{Aut}(\mathbb{O})$ acts on $\mathbb{O}$ preserving the inner product and the real subspace, so it acts on $\mathrm{Im}(\mathbb{O})$. As established in Proposition 5.9, G_2 acts transitively on ordered orthonormal pairs in $\mathrm{Im}(\mathbb{O})$ — the simple transitivity of G_2 on basic triples (Baez, 2002, §4.1), restricted to the first two frame vectors (cf. Harvey 1990 for the Stiefel-manifold formulation). The aphanic conditions $\|x\| = \|y\|$, $\langle x, y\rangle = 0$ say exactly that (x, y) is an unordered orthonormal pair (after rescaling), so G_2 acts transitively on the projectivized aphanic locus. The stabilizer of the basepoint (e_1, e_2) in G_2 is the subgroup fixing both e_1 and e_2; this is the $\mathrm{SU}(2) \subset G_2$ acting on the quaternionic complement $\mathbb{O} \ominus \mathbb{H}(e_1, e_2)$ in the standard way. Therefore $\mathcal{V} \cong G_2/\mathrm{SU}(2)$. □

The image of the kernel map $\Phi : \mathcal{V} \to \mathrm{Gr}(4, \mathbb{S})$ is diffeomorphic to $G_2/U(2)$, the twistor space of S^6 (Bryant, 1982). The fibration $G_2/\mathrm{SU}(2) \to G_2/U(2)$ has fiber S^1, captured geometrically by the rotation of the (e_1, e_2) frame.

6.2 G_2-Equivariance of the Kernel Bundle

Lemma 6.2 (G_2-Invariance of $\mathcal{K}$)**.** *The doubled G_2-action on $\mathbb{S}$, $\widetilde{D} := \mathrm{diag}(D, D)$ for $D \in G_2 \subset \mathrm{SO}(\mathbb{O})$, satisfies $\widetilde{D} \in \mathrm{Aut}(\mathbb{S})$. Consequently the kernel bundle $\mathcal{K}$ is G_2-equivariant: for every $g \in G_2$ and $p \in \mathcal{V}$,*

$$\widetilde{g} \cdot \ker L_p \;=\; \ker L_{\widetilde{g \cdot p}}.$$

Proof. For $g \in G_2$, $g(uv) = g(u)g(v)$ in $\mathbb{O}$, $g(\bar{u}) = \overline{g(u)}$, and $\|g(u)\| = \|u\|$. Applying the recursive product (1) and using these, $\widetilde{g}((a, b)(c, d)) = (\widetilde{g}(a, b))(\widetilde{g}(c, d))$. Hence $\widetilde{g} \in \mathrm{Aut}(\mathbb{S})$. For any aphanic p and $z \in \ker L_p$, $L_p(z) = pz = 0$, so $\widetilde{g}(p) \cdot \widetilde{g}(z) = \widetilde{g}(pz) = 0$, i.e., $\widetilde{g}(z) \in \ker L_{\widetilde{g}(p)}$. The reverse inclusion follows from $\widetilde{g}^{-1}$, completing the proof. □

Theorem 6.3 (Fiber Representation Theorem)**.** *The stabilizer $\mathrm{SU}(2) \subset G_2$ of the basepoint $p_0 = e_1 + e_2 \cdot \ell$ acts on $\ker L_{p_0}$ by the fundamental (spin-$\frac{1}{2}$) representation. Under this action, $\ker L_{p_0} \cong \mathbb{H}$ as an $\mathrm{SU}(2) \cong \mathrm{Sp}(1)$-module.*

Sketch. The Lie subalgebra $\mathfrak{s} = \{D \in \mathfrak{g}_2 : De_1 = De_2 = 0\}$ has dimension 3 (rank-11 constraint in 14-dimensional $\mathfrak{g}_2$). After an exact rational normalization (certified in `op_six_anchors_A`; the earlier numerical Gram–Schmidt is superseded), three generators $S_0, S_1, S_2 \in \mathfrak{s}$ satisfy $[S_i, S_j] = 6\,\varepsilon_{ijk} S_k$. Lifting to $\mathbb{S}$ via $\widetilde{S}_i = \mathrm{diag}(S_i, S_i)$ and restricting to $K_0 := \ker L_{p_0}$ gives operators on $\mathbb{H}$ which, after rescaling $T_a = S_a/6$, satisfy $T_a^2 = -\frac{1}{4}\mathrm{id}_{K_0}$, $\{T_a, T_b\} = 0$ for $a \neq b$, $[T_a, T_b] = T_c$ cyclically. This is the spin-$\frac{1}{2}$ representation of $\mathfrak{su}(2)$. The quadratic Casimir $C_2 = \sum_a T_a^2 = -\frac{3}{4}\mathrm{id}_{K_0}$ yields $j(j+1) = 3/4$, $j = 1/2$, confirming irreducibility. □

The lemma and the fiber representation theorem together identify $\mathcal{K} \to \mathcal{V}$ with the associated bundle $G_2 \times_{\mathrm{SU}(2)} \mathbb{H}$, where $\mathrm{SU}(2)$ acts on $\mathbb{H}$ by left multiplication (its fundamental two-dimensional complex representation, equivalently the four-dimensional real representation $\mathrm{Sp}(1) \curvearrowright \mathbb{H}$).

6.3 Bundle Nontriviality

Theorem 6.4 (Bundle Nontriviality / Aphanic Twist)**.** *The kernel bundle $\mathcal{K} = G_2 \times_{\mathrm{SU}(2)} \mathbb{H}$ over $\mathcal{V} = G_2/\mathrm{SU}(2)$ is topologically nontrivial. The characteristic class $[\mathcal{K}] \in H^*(\mathcal{V}; \mathbb{Z}/2)$ is nonzero.*

Proof. The cohomology of $\mathcal{V} = G_2/\mathrm{SU}(2)$ is computed from the Serre spectral sequence of the fibration $S^5 \to V_2(\mathbb{R}^7) \to S^6$ (where $V_2(\mathbb{R}^7) \cong \mathcal{V}$ is the Stiefel manifold of orthonormal 2-frames in $\mathbb{R}^7$):

$$H^k(\mathcal{V}; \mathbb{Z}) \;=\; \begin{cases} \mathbb{Z} & k = 0, 11, \\ \mathbb{Z}/2 & k = 6, \\ 0 & \text{otherwise.} \end{cases}$$

All primary characteristic classes ($w_1, \dots, w_4$, p_1, Euler) vanish for dimensional reasons: they live in H^k for $k \le 4$, all of which are zero. Hence $\mathcal{K}$ is stably trivial, and the only possible nonzero invariant is a secondary class in $H^6(\mathcal{V}; \mathbb{Z}/2) = \mathbb{Z}/2$.

To detect this class, restrict $\mathcal{K}$ to the fiber S^5 of the Serre fibration. The classifying map of S^5 as the principal $\mathrm{SU}(2)$-bundle $\mathrm{SU}(2) \hookrightarrow \mathrm{SU}(3) \to S^5$ gives, via the long exact sequence of homotopy groups,

$$\pi_5(\mathrm{SU}(3)) \;\to\; \pi_5(S^5) \;\xrightarrow{\partial}\; \pi_4(\mathrm{SU}(2)) \;\to\; \pi_4(\mathrm{SU}(3)) \;=\; 0.$$

Since $\pi_5(S^5) = \mathbb{Z}$, $\pi_4(\mathrm{SU}(2)) = \mathbb{Z}/2$, and the target $\pi_4(\mathrm{SU}(3)) = 0$, the connecting homomorphism ∂ is surjective. The generator of $\pi_5(S^5)$ maps to the nontrivial element of $\pi_4(\mathrm{SU}(2))$, so the principal $\mathrm{SU}(2)$-bundle $\mathrm{SU}(3) \to S^5$ is the unique nontrivial $\mathrm{SU}(2)$-bundle on S^5. Its associated bundle $\mathrm{SU}(3) \times_{\mathrm{SU}(2)} \mathbb{H}$ is the unique nontrivial quaternionic line bundle on S^5, and this is exactly the restriction of $\mathcal{K}$ to the fiber. Therefore $\mathcal{K}$ is nontrivial. □

Remark 6.5 (Scope of the topological statement). Theorem 6.4 asserts nontriviality, detected through the $\mathrm{SU}(2)$-structure of the fiber restriction, and nothing more. It does not classify rank-4 real bundles over $\mathcal{V}$: a full classification involves the obstructions in $H^6(\mathcal{V}; \pi_4(\mathrm{SO}(4))) = \mathbb{Z}/2 \oplus \mathbb{Z}/2$ together with the top cell of $\mathcal{V}$, and is left open here.

Theorem 6.6 (Berry Phase Theorem (Berry, 1984; Simon, 1983))**.** *The canonical (Nomizu) connection (Kobayashi and Nomizu, 1963; Nomizu, 1954) on $\mathcal{K} \to \mathcal{V}$ is non-flat. There exist closed loops $\Gamma \subset \mathcal{V}$ around which the parallel transport on $\mathcal{K}$ is the operator $-\mathrm{id}_{\mathbb{H}}$.*

Sketch. Consider the loop $\Gamma_A : \theta \mapsto ((\cos\theta)e_1 + (\sin\theta)e_3) + e_2 \cdot \ell$, $\theta \in [0, 2\pi]$. This is a closed geodesic in $\mathcal{V}$. Direct integration of the Nomizu connection 1-form along Γ_A, using the explicit form of the connection in the orthonormal frame, gives holonomy $-\mathrm{id}$. This is now certified in closed form, with no integration: writing $\Gamma_A(\theta) = \exp(\theta X) \cdot p_0$ for the unique $X \in \mathfrak{g}_2$ with $Xe_1 = e_3$, $Xe_3 = -e_1$, $Xe_2 = 0$ lying in the reductive complement $\mathfrak{m}$ of $\mathfrak{s}$, the curve $\theta \mapsto \exp(\theta X)$ is the horizontal lift of Γ_A for the canonical connection (Kobayashi and Nomizu, 1963), so the holonomy is the action of $h := \exp(2\pi X)$ on the fibre. Exactly, X splits $\mathrm{Im}(\mathbb{O})$ as $\langle e_2 \rangle \oplus \langle e_1, e_3 \rangle \oplus \langle e_4, \dots, e_7 \rangle$ with $X^2 = 0$, $-\mathrm{Id}$ and $-\frac{1}{4}\mathrm{Id}$ respectively; the half-integer frequency on the last block gives $h = \mathrm{diag}(+1 \text{ on } \{1, e_1, e_2, e_3\}, -1 \text{ on } \{e_4, \dots, e_7\})$ in closed form. One checks exactly that $h \in \mathrm{Aut}(\mathbb{O})$, that h fixes p_0 (so the loop closes), and that $\widetilde{h} = \mathrm{diag}(h, h)$ preserves K with $\widetilde{h}|_K = -\mathrm{id}$.

Because $\mathcal{V}$ is simply connected, every loop bounds a disc and the holonomy group coincides with the restricted holonomy group; by the Ambrose–Singer theorem (Kobayashi and Nomizu, 1963) it is generated by curvature transported over a filling disc. Non-flatness is therefore

a consequence of the computation rather than a separate hypothesis: a flat connection on a simply connected base has trivial holonomy, whereas Γ_A transports by $-\mathrm{id} \neq \mathrm{id}$. Other geodesic loops may still have trivial holonomy (e.g., loops in the S^5 fiber not crossing the antipode); the holonomy of each is fixed by the curvature over its own filling disc, and — $\mathcal{V}$ being simply connected — no homotopy-class bookkeeping enters.

The class of Theorem 6.4 is not what Γ_A measures. That class lives in $H^6(\mathcal{V}; \mathbb{Z}/2)$ and is registered by the clutching function of $\mathcal{K}$ over the equatorial $S^4 \subset S^5$ through $\pi_4(\mathrm{SU}(2)) = \mathbb{Z}/2$; a 1-cycle cannot detect it. The $-\mathrm{id}$ holonomy of Γ_A and the nonvanishing $\mathbb{Z}/2$ class are independent facts about $\mathcal{K}$ — the first metric-geometric, the second topological. □

Remark 6.7 (Status). Theorem 6.3 is PROVEN, with no residual floating-point dependence: $\dim \mathfrak{g}_2 = 14$ and $\dim \mathfrak{s} = 3$ are certified by exact rational row reduction paired with exhibited nonzero minors (bounding each rank above and below), the normalization of the T_a is exact-rational, and the Casimir is the exact scalar $-\frac{3}{4}\,\mathrm{id}$. Irreducibility is certified over $\mathbb{R}$, of quaternionic type — the complexification is two copies of the spin-$\frac{1}{2}$ representation. Theorem 6.4 is PROVEN (Serre spectral sequence, long exact sequence of homotopy groups). Theorem 6.6 is PROVEN: the numerical integration that formerly carried it is superseded by the exact closed-form evaluation of $\exp(2\pi X)$ given above. Exactly the following are certified in exact arithmetic: that X exists in $\mathfrak{g}_2$, is *unique*, and is Frobenius-orthogonal to $\mathfrak{s}$; that $Xe_1 = e_3$, $Xe_2 = 0$, $Xe_3 = -e_1$; that $\mathrm{Im}(\mathbb{O})$ splits X-invariantly as $\langle e_2\rangle \oplus \langle e_1, e_3\rangle \oplus \langle e_4, \ldots, e_7\rangle$ with X^2 equal to 0, $-\mathrm{Id}$ and $-\frac{1}{4}\mathrm{Id}$ on the three blocks; that the resulting $h = \exp(2\pi X)$ lies in $\mathrm{Aut}(\mathbb{O})$ (checked on all 64 basis pairs) and satisfies $h^2 = \mathrm{Id}$; that h fixes p_0, so Γ_A closes; that $\widetilde{h}$ preserves $\mathcal{K}$ by exact containment; and that $\widetilde{h}|_{\mathcal{K}} = -\mathrm{id}$. Exactly one ingredient is non-computational, and it is a standard fact rather than a new claim: on a reductive homogeneous space, $\exp(tX)$ is horizontal for the canonical connection precisely when $X \in \mathfrak{m}$ (Kobayashi and Nomizu, 1963). No floating-point quantity enters the theorem or its proof.

7 Algebraic Identification of the Holonomy Factors

It is established in prior work that the natural connection on the kernel sub-bundle $K \to \mathcal{P}_{\mathrm{aph}}$ has holonomy algebra $\mathfrak{u}(2) = \mathfrak{u}(1) \oplus \mathfrak{su}(2)$ and holonomy group $U(2) = (U(1) \times SU(2))/\mathbb{Z}_2$. That statement is intrinsic: it identifies the Lie algebra up to isomorphism but does not locate its factors inside the ambient algebra $\mathfrak{so}(K)$ of skew-symmetric endomorphisms of a single fiber. The present section closes that gap. Fixing the base point $p_0 = e_1 + e_2 \cdot \ell$, we give both factors explicit generators inside the sedenion multiplication structure.

7.1 Setup

Throughout this section, all sedenion multiplications are the canonical recursive Cayley–Dickson product

$$(a, b)(c, d) \;=\; (ac - \bar{d}b,\; da + b\bar{c}).$$

Let $K := \ker L_{p_0}$, a 4-dimensional real subspace of $\mathbb{S}$; the standard computation yields the orthonormal basis

$$k_0 = \tfrac{1}{\sqrt{2}}(e_7 + e_{12}), \quad k_1 = \tfrac{1}{\sqrt{2}}(-e_6 + e_{13}), \quad k_2 = \tfrac{1}{\sqrt{2}}(e_5 + e_{14}), \quad k_3 = \tfrac{1}{\sqrt{2}}(-e_4 + e_{15}). \tag{29}$$

Identify $K \cong \mathbb{H}$ by $(a_0, a_1, a_2, a_3) \mapsto a_0 + a_1\,\mathbf{i} + a_2\,\mathbf{j} + a_3\,\mathbf{k}$. Under this identification, the ambient Lie algebra decomposes as

$$\mathfrak{so}(K) \;=\; \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R,$$

where $\mathfrak{su}(2)_L$ (resp. $\mathfrak{su}(2)_R$) consists of left (resp. right) multiplication by imaginary quaternions on $\mathbb{H}$. These two subalgebras centralize each other; their direct sum exhausts $\mathfrak{so}(4)$.

Let $G_2 = \mathrm{Aut}(\mathbb{O})$ act on $\mathbb{S}$ by the doubled representation $\widetilde{D} = \mathrm{diag}(D, D)$ with $D \in \mathrm{SO}(\mathbb{O})$, and let $\mathfrak{g}_2 \subset \mathfrak{so}(\mathbb{O})$ denote its Lie algebra, realized concretely as the span of the octonion derivations $D_{a,b} = [L_a, L_b] + [L_a, R_b] + [R_a, R_b]$, $a, b \in \mathrm{Im}(\mathbb{O})$, with $\dim \mathfrak{g}_2 = 14$.

7.2 The stabilizer inside $\mathfrak{g}_2$

Proposition 7.1. *The subalgebra*

$$\mathfrak{s} \;:=\; \{\, D \in \mathfrak{g}_2 \;:\; De_1 = 0,\; De_2 = 0 \,\}$$

is three-dimensional, and it generates the copy of $\mathfrak{su}(2)$ whose quotient produces $\mathcal{P}_{\mathrm{aph}} \cong G_2/SU(2)$.

Proof. Express each $D \in \mathfrak{g}_2$ in an orthonormal basis of $\mathfrak{g}_2$ (obtained by Gram–Schmidt on the 49-dimensional ambient space $\mathrm{Mat}_7(\mathbb{R})$ applied to a spanning set of derivations). The two conditions $De_1 = 0$ and $De_2 = 0$ give 14 linear equations in 14 unknowns. Exact rational row reduction gives rank 11, hence $\dim \mathfrak{s} = 3$. The rank is certified in exact arithmetic in both directions: fraction-free elimination over $\mathbb{Q}$ bounds it above, and an exhibited nonzero 11×11 minor bounds it below. The prerequisite $\dim \mathfrak{g}_2 = 14$ is certified the same way, by solving the Leibniz system over all 64 basis pairs rather than assuming that a chosen spanning set of derivations exhausts $\mathfrak{g}_2$. The identification $\mathfrak{s} \cong \mathfrak{su}(2)$ follows from item (i) of Theorem 7.2 below. □

Fix generators $S_0, S_1, S_2 \in \mathfrak{s}$ (any basis will do). Define their lifts and fiber restrictions

$$J_a \;:=\; \mathrm{diag}(S_a, S_a)\,\big|_K, \qquad a \in \{0, 1, 2\}. \tag{30}$$

Each $J_a \in \mathfrak{so}(K)$ by construction: the lift preserves K *exactly* (certified as exact containment over $\mathbb{Q}$, with no leakage tolerance), since K is G_2-invariant; cf. Lemma 6.2.

7.3 The decomposition theorem

Let $G_1, G_2, G_3, G_4 \in \mathfrak{so}(K)$ denote the four curvature generators of that prior work (the central generator $G_1 = J_\Delta$ and the Gram–Schmidt basis $\{G_2, G_3, G_4\}$ of the non-abelian factor). For $v \in \mathrm{Im}(\mathbb{S})$, write $R_v|_K$ for the restriction to K of right-multiplication by v on $\mathbb{S}$ (when K is R_v-invariant).

Theorem 7.2 (Holonomy Decomposition)**.** *Under the identification $K \cong \mathbb{H}$ induced by (29), the holonomy algebra of the natural connection on $K \to \mathcal{P}_{\mathrm{aph}}$ at p_0 decomposes as*

$$\mathfrak{hol} \;=\; \mathfrak{u}(2) \;=\; \mathfrak{u}(1)_R \oplus \mathfrak{su}(2)_L, \tag{31}$$

where:

(i) *Setting $I_a := J_a/3$ from (30), one has $I_a^2 = -\mathrm{Id}$, $\{I_a, I_b\} = 0$ for $a \neq b$, and $[I_0, I_1] = 2\, I_2$ (with cyclic permutations). Consequently $\mathrm{span}_{\mathbb{R}}\{J_0, J_1, J_2\} = \mathfrak{su}(2)_L$.*

(ii) $\mathrm{span}_{\mathbb{R}}\{G_2, G_3, G_4\} = \mathfrak{su}(2)_L$*, with explicit change of basis*

$$G_2 = \tfrac{1}{6}\, J_1, \qquad G_3 = \tfrac{1}{6}\, J_0, \qquad G_4 = -\tfrac{1}{6}\, J_2.$$

(iii) $[G_1, J_a] = 0$ *for $a \in \{0, 1, 2\}$; hence G_1 lies in the centralizer of $\mathfrak{su}(2)_L$ in $\mathfrak{so}(K)$, which is $\mathfrak{su}(2)_R$.*

(iv) *The right-multiplications $R_{e_1}|_K$ and $R_{e_2}|_K$ are well-defined elements of $\mathfrak{so}(K) \cap \mathfrak{su}(2)_R$, and*

$$G_1 \;=\; -\tfrac{1}{4}\,[\,R_{e_1}|_K,\; R_{e_2}|_K\,].$$

In particular, $\mathfrak{u}(1)_R = \mathbb{R}{\cdot}G_1$ is the Lie closure of the right-multiplication pair $\{R_{e_1}|_K,\, R_{e_2}|_K\}$ in $\mathfrak{so}(K)$.

Proof. All identities below are verified in *exact rational arithmetic* by the `op_six_anchors_A` certification probe — every residual is the identically zero matrix over $\mathbb{Q}$, and every dimension count is a fraction-free row reduction paired with an exhibited nonzero minor. No floating-point computation is load-bearing in any statement of this theorem. Here we give the structural argument.

(i) A direct matrix computation from (30) yields $J_a^2 = -9\,\mathrm{Id}$, $\{J_a, J_b\} = 0$ $(a \neq b)$, and $[J_0, J_1] = 6\,J_2$ with cyclic permutations. Rescaling $I_a := J_a/3$ gives the standard imaginary-quaternion relations $I_a^2 = -\mathrm{Id}$, $\{I_a, I_b\} = 0$, $[I_0, I_1] = 2\,I_2$. Three elements of $\mathfrak{so}(4)$ satisfying these relations necessarily generate $\mathfrak{su}(2)_L$ under the identification $K \cong \mathbb{H}$; indeed, left-multiplication $L^{\mathbb{H}}_{\mathbf{i}}$, $L^{\mathbb{H}}_{\mathbf{j}}$, $L^{\mathbb{H}}_{\mathbf{k}}$ on $\mathbb{H}$ satisfy exactly these relations and span $\mathfrak{su}(2)_L$ by definition.

(ii) Project each G_m $(m = 2, 3, 4)$ onto $\mathrm{span}\{J_0, J_1, J_2\}$ in the Frobenius inner product; the residuals are zero, giving the stated change of basis. Equality of the two three-dimensional subspaces follows.

(iii) Direct commutator computation: $[G_1, J_a] = 0$ for $a = 0, 1, 2$, exactly over $\mathbb{Q}$. The centralizer of $\mathrm{span}\{J_a\}$ in $\mathfrak{so}(K)$ is moreover certified to have dimension exactly 3 by exact row reduction. Since $\mathfrak{so}(K) = \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ with each factor equal to the other's centralizer (a standard fact about $\mathfrak{so}(4)$), the conclusion is immediate.

(iv) That R_{e_1} and R_{e_2} preserve K at p_0 is a direct verification, certified as *exact* containment (zero leakage, not a bounded one); this uses only that e_1, e_2 lie in the quaternionic subalgebra $\mathbb{H}$ spanned by $\{1, e_1, e_2, e_3\}$ that appears in the Cayley–Dickson construction of p_0. A direct commutator computation gives

$$-\tfrac{1}{4}\,[R_{e_1}|_K,\, R_{e_2}|_K] \;=\; G_1,$$

as an exact identity over $\mathbb{Q}$, with $G_1^2 = -\frac{1}{4}\,\mathrm{Id}$ pinning the normalization exactly. The Lie-closure statement follows: the pair $\{R_{e_1}|_K,\, R_{e_2}|_K\}$ spans only a two-dimensional subspace of $\mathfrak{su}(2)_R$, but their bracket supplies the third dimension, and the line $\mathbb{R} \cdot [R_{e_1}|_K, R_{e_2}|_K]$ is precisely $\mathfrak{u}(1)_R$. □

7.4 Remarks

Remark 7.3 (Origin of each factor). Theorem 7.2 supplies each summand of $\mathfrak{u}(2)$ with a distinct algebraic origin inside $\mathbb{S}$:

- The non-abelian factor $\mathfrak{su}(2)_L$ is the restriction to K of the lifted action of the $SU(2) \subset G_2$ stabilizer of p_0. The symmetry quotiented out to form the aphanic locus $G_2/SU(2)$ reappears inside the fiber as the non-abelian rotation.
- The abelian factor $\mathfrak{u}(1)_R$ is not generated by a single elementary right-multiplication on K; it is generated by the Lie bracket of two such right-multiplications by orthogonal units of the distinguished quaternion subalgebra $\langle 1, e_1, e_2, e_3\rangle$. The Berry phase generator is, algebraically, a commutator of right-actions.

Remark 7.4 (Reshaping of an earlier conjecture). Earlier working versions of this framework conjectured that the non-abelian factor would identify with right-multiplication by imaginary quaternions on K (*i.e.*, with $\mathfrak{su}(2)_R$). Theorem 7.2 shows the opposite: the non-abelian factor is left-multiplication, while the abelian factor is a distinguished line inside the right-multiplication algebra. The geometric content of the two factors is therefore reversed relative to that conjecture.

Remark 7.5 (Reproducibility). Every claim in the proof is certified in exact rational arithmetic by a single self-contained C++17 probe (`op_six_anchors_A.cpp`) with no external dependencies, superseding the earlier floating-point verification (`holonomy_u2_identification.cpp`). Integer arithmetic throughout, with 128-bit internals and hard overflow checks; every residual is the identically zero matrix and every dimension count is a fraction-free row reduction paired with an exhibited nonzero minor, so each rank is bounded above and below in exact arithmetic. There is no leakage tolerance anywhere: subspace invariance is certified as exact containment. Nothing in this section rests on floating point.

Remark 7.6 (No physics claim). The Lie group $U(2) = (U(1) \times SU(2))/\mathbb{Z}_2$ is, up to isomorphism, the electroweak gauge group of the Standard Model. The results here make no physical claim; they are strictly mathematical statements about the holonomy of a natural connection on the sedenion kernel sub-bundle. What the sedenion algebra has produced, without physical input, is the Lie group $U(2)$. What it has not done is supply a metric, a bundle of Dirac spinors, a Yukawa sector, or any of the structure required to interpret this as a physical gauge theory. That interpretation, if it is warranted at all, is a separate question.

8 Algebraic $\mathbf{R}_\ell$ Centrality

Section 7 identified the abelian factor $\mathfrak{u}(1)_R \subset \mathfrak{u}(2)$ inside the ambient sedenion multiplication algebra at the basepoint $p_0 = e_1 + e_2 \cdot \ell$. The identification was *local*: it placed the generator $G_1 = -\frac{1}{4}[R_{e_1}|_K, R_{e_2}|_K]$ inside $\mathfrak{su}(2)_R$ at the single fiber over p_0. The Three-Way Identity strengthened this to the *Three-Way Identity*, $2\,G_1 = R_\ell|_{K_{p_0}} = \sqrt{2}\,B_3$, which showed that three independent constructions of the U(1) center—Berry, CD algebra, frame-free Hodge—collapse to a single line at p_0.

This section answers the question carried from that identity and only partly addressed by the earlier orbit-uniform numerical statement (which certified $[R_\ell, \pi_p] = 0$ on $\mathcal{P}_{\text{aph},1}$ at machine precision across $N = 64$ random samples but offered no algebraic derivation): *is the U(1) generator a fiberwise restriction of a single p-independent operator on $\mathbb{S}$, and is the centrality identity algebraic rather than numerical?*

The answer is yes, with a sharper statement than the kernel-projector version: the centrality identity holds for the *full Gram operator* $G_p = L_p^\top L_p$ across the entire imaginary locus, of which $\mathcal{P}_{\text{aph}}$ is the kernel-bearing sublocus, and the proof is a five-stage reduction in the Cayley–Dickson block algebra.

8.1 Statement

Theorem 8.1 (Algebraic R_ℓ Centrality)**.** *Let $\mathbb{S} = \mathbb{O} \oplus \mathbb{O} \cdot \ell$ be the sedenion algebra under canonical recursive Cayley–Dickson doubling $(a, b)(c, d) = (ac - \bar{d}b,\ da + b\bar{c})$, with $\ell = e_8$. Let $p = (x, y)$ with $x, y \in \mathrm{Im}(\mathbb{O})$, L_p left-multiplication by p, $G_p := L_p^\top L_p$ the Gram operator, and R_ℓ right-multi plication by ℓ. Then*

$$[R_\ell,\, G_p] \;=\; 0 \qquad \text{on } \mathbb{R}^{16}. \tag{32}$$

Corollary 8.2 (Fiberwise centrality)**.** *For every $p \in \mathcal{P}_{\text{aph},1}$ the orthogonal projector π_p onto $\ker(L_p)$ satisfies $[R_\ell, \pi_p] = 0$.*

Proof of Corollary 8.2. On the aphanic locus the Gram operator G_p has spectrum $\{0^4, 2^8, 4^4\}$ (the 4-8-4 Spectral Theorem); the three eigenvalues are distinct, so each eigenprojector is a polynomial in G_p given by Lagrange interpolation:

$$\pi_p \;=\; \prod_{\mu \in \{2,4\}} \frac{G_p - \mu I}{0 - \mu} \;=\; \tfrac{1}{8}(G_p - 2I)(G_p - 4I).$$

Any operator commuting with G_p commutes with every polynomial in G_p, hence with π_p. Theorem 8.1 gives $[R_\ell, G_p] = 0$, so $[R_\ell, \pi_p] = 0$. □

8.2 Strict strengthening over the fiberwise statement

Theorem 8.1 is strictly stronger than the numerical that statement in two independent senses.

Remark 8.3 (Locus). The fiberwise statement holds on the aphanic locus $\mathcal{P}_{\text{aph},1}$ (the unit-norm aphanic observers, an 11-dimensional homogeneous space $G_2/SU(2)$). Theorem 8.1 holds on the *imaginary locus* $\{p = (x, y) \in \mathbb{S} : x, y \in \mathrm{Im}(\mathbb{O})\}$, a 14-dimensional real submanifold of $\mathbb{S}$ of which the aphanic locus is a strict, codimension-3 subset. The aphanic conditions $\|x\| = \|y\|$ and $\langle x, y\rangle = 0$ play *no role* in the proof.

Remark 8.4 (Operator). It is an identity on π_p, the rank-4 spectral projector of G_p. Theorem 8.1 is an identity on the full operator G_p, of which π_p is one of three spectral projectors (for the eigenvalues 0, 2, 4). The full identity contains the kernel-restricted version as a corollary.

8.3 Proof

Throughout this subsection let

$$K \;:\; \mathbb{O} \to \mathbb{O}, \quad K(u) := \bar{u}$$

denote octonion conjugation, with matrix $K = \mathrm{diag}(1, -1, -1, \ldots, -1) \in \mathbb{R}^{8\times 8}$ in the standard basis $\{1, e_1, \ldots, e_7\}$ of $\mathbb{O}$. Define the *associator-action operator*

$$A_{x,y} : \mathbb{O} \to \mathbb{O}, \qquad A_{x,y}(u) := [x, u, y] := (xu)y - x(uy).$$

The proof proceeds in five stages, each a self-contained algebraic identity verified at machine precision in the C++17 probe `op_rell_centrality_proof.cpp` on $1 + 64 + 64 = 129$ samples spanning p_0, the aphanic orbit, and the imaginary-non-aphanic regime (Section 8.4).

Stage 1: block decomposition of L_p and G_p

Identify $\mathbb{S}$ with $\mathbb{O} \oplus \mathbb{O}$ via $(u, v) \leftrightarrow u + v \cdot \ell$. Direct application of the Cayley–Dickson product gives

$$L_p \;=\; \begin{pmatrix} L_x & -R_y K \\ L_y K & R_x \end{pmatrix}, \qquad R_\ell \;=\; \begin{pmatrix} 0 & -I_8 \\ I_8 & 0 \end{pmatrix}. \tag{33}$$

The composition-algebra identities $L_{\bar{a}} L_a = R_{\bar{a}} R_a = \|a\|^2 I$ in $\mathbb{O}$, together with $K^2 = I$, $K^\top = K$, $L_a^\top = L_{\bar{a}}$, $R_a^\top = R_{\bar{a}}$, give

$$G_p \;=\; L_p^\top L_p \;=\; \begin{pmatrix} \|p\|^2\, I_8 & B \\ B^\top & \|p\|^2\, I_8 \end{pmatrix}, \qquad B \;=\; -L_{\bar{x}} R_y K + K L_{\bar{y}} R_x. \tag{34}$$

For $x, y \in \mathrm{Im}(\mathbb{O})$ ($\bar{x} = -x$, $\bar{y} = -y$) this specializes to

$$B \;=\; L_x R_y K - K L_y R_x. \tag{35}$$

Stage 2: reduction to skewness

Direct block multiplication of the matrices in (33) and (34) yields

$$[R_\ell,\, G_p] \;=\; \begin{pmatrix} -(B + B^\top) & 0 \\ 0 & B + B^\top \end{pmatrix}. \tag{36}$$

Hence $[R_\ell, G_p] = 0 \iff B$ is skew-symmetric.

Stage 3: associator identity for the cross-block

Apply B in (35) to $u \in \mathbb{O}$:

$$B(u) \;=\; x \cdot (\bar{u}\, y) \;-\; \overline{y(ux)}.$$

The Cayley–Dickson conjugation is an anti-homomorphism, $\overline{ab} = \bar{b}\bar{a}$, so $\overline{y(ux)} = \overline{ux} \cdot \bar{y} = (\bar{x}\bar{u})\bar{y}$. For $x, y \in \mathrm{Im}(\mathbb{O})$ this is $(-x\bar{u})(-y) = (x\bar{u})\, y$, giving

$$B(u) \;=\; x(\bar{u}\, y) - (x\bar{u})\, y \;=\; -\big[(x\bar{u})y - x(\bar{u}y)\big] \;=\; -[x,\, \bar{u},\, y] \;=\; -A_{x,y}(Ku). \tag{37}$$

Thus, as 8×8 real matrices,

$$B \;=\; -A_{x,y} \circ K. \tag{38}$$

The minus sign is essential and traces to the parity of $\bar{y} = -y$ in $\overline{y(ux)}$.

Stage 4: $A_{x,y}$ is skew on $\mathbb{O}$ for $x, y \in \mathrm{Im}(\mathbb{O})$

Stages 4 and 5 are written at the octonion level, since that is the level at which the sedenion cross-block lives. The ingredients they use — flexibility, the inner-product duality $\langle uv, w\rangle = \langle v, \bar{u}w\rangle$, and the conjugation anti-homomorphism property — hold at every Cayley–Dickson level (Table 1), so the arguments transpose verbatim one level up; this is what Section 9 makes explicit.

In a normed composition algebra, $\langle ab, c\rangle = \langle b, \bar{a}c\rangle = \langle a, c\bar{b}\rangle$. Applying both forms,

$$\begin{aligned}\langle [x,u,y],\, v\rangle &= \langle (xu)y, v\rangle - \langle x(uy), v\rangle \\ &= \langle u,\, \bar{x}(v\bar{y})\rangle - \langle u,\, (\bar{x}v)\bar{y}\rangle \\ &= -\langle u,\, [\bar{x}, v, \bar{y}]\rangle.\end{aligned}$$

For $x, y \in \mathrm{Im}(\mathbb{O})$, $[\bar{x}, v, \bar{y}] = [-x, v, -y] = [x, v, y]$, hence

$$A_{x,y}^{\top} \;=\; -A_{x,y}. \tag{39}$$

Stage 5: K commutes with $A_{x,y}$ for $x, y \in \mathrm{Im}(\mathbb{O})$

By the anti-homomorphism property,

$$\overline{[x,u,y]} \;=\; \overline{(xu)y} - \overline{x(uy)} \;=\; \bar{y}(\bar{u}\bar{x}) - (\bar{y}\bar{u})\bar{x} \;=\; -[\bar{y}, \bar{u}, \bar{x}].$$

For $x, y \in \mathrm{Im}(\mathbb{O})$, $-[\bar{y}, \bar{u}, \bar{x}] = -[-y, \bar{u}, -x] = -[y, \bar{u}, x] = [x, \bar{u}, y]$ by the alternation of the associator under transposition of outer arguments. Therefore

$$K\, A_{x,y}(u) \;=\; \overline{[x,u,y]} \;=\; [x, \bar{u}, y] \;=\; A_{x,y}(Ku), \tag{40}$$

i.e., $[K, A_{x,y}] = 0$.

Closure

Combining (38), (39), (40):

$$B^{\top} \;=\; -(A_{x,y}K)^{\top} \;=\; -K^{\top}A_{x,y}^{\top} \;=\; -K(-A_{x,y}) \;=\; KA_{x,y} \;=\; A_{x,y}K \;=\; -B.$$

So B is skew, and (36) yields $[R_\ell, G_p] = 0$. □

8.4 Numerical certification

Each of Stages 1–5 is a matrix identity over $\mathbb{R}$. The probe `op_rell_centrality_proof.cpp` certifies all five at machine precision on three sample sets:

- the basepoint $p_0 = e_1 + e_2 \cdot \ell$ (every residual identically zero, $0 \cdot 10^0$);
- $N = 64$ random aphanic observers $p \in \mathcal{P}_{\text{aph},1}$ (worst residual $2.34 \cdot 10^{-15}$, Stage 2 commutator at the 16×16 level);
- $N = 64$ random imaginary-non-aphanic observers ($x, y \in \mathrm{Im}(\mathbb{O})$ with generic norms and inner product; worst residual $2.33 \cdot 10^{-14}$).

All residuals are $\leq 10^{-12}$, the declared tolerance. The third sample set is the operative test for the strict strengthening claimed in Section 8.2: it lies outside $\mathcal{P}_{\text{aph}}$ but inside the proof's scope.

8.5 Remarks

Remark 8.5 (Why the imaginary locus is the natural domain). The proof uses $\bar{x} = -x$ and $\bar{y} = -y$ in two essential places: in the simplification of $\overline{y(ux)}$ to $(x\bar{u})y$ (Stage 3, equation (37)) and in collapsing $[\bar{x}, v, \bar{y}] = [x, v, y]$ (Stage 4, justifying $A^\top = -A$). Real components of x or y would propagate through both reductions and break the skew-symmetry of B. The proof tells us, in passing, that the imaginary locus is the maximal subset of $\mathbb{S}$ on which R_ℓ is algebraically central with respect to the Gram operator.

Remark 8.6 (Reframing of the Section 6 U(1) generator). Theorem 7.2 gave the U(1) center at p_0 as the commutator $-\frac{1}{4}[R_{e_1}|_{K_{p_0}}, R_{e_2}|_{K_{p_0}}] = G_1$. That construction is basepoint-bound: R_{e_1} and R_{e_2} stop preserving K_p as p moves off the quaternionic distinguished subalgebra. Theorem 8.1 replaces the commutator construction with a single ambient operator $R_\ell = R_{e_8}$ that is central with respect to G_p at every imaginary p. Combined with the Three-Way Identity ($2G_1 = R_\ell|_{K_{p_0}}$ at p_0), the upgrade is: the U(1) generator is a single global matrix on $\mathbb{S}$, not a fiberwise commutator.

Remark 8.7 (Reproducibility). The five-stage decomposition is realized in a single self-contained C++17 probe (`op_rell_centrality_proof.cpp`) with no external dependencies, building under both MSVC and GCC; runtime is sub-second. The probe emits per-stage worst-case residuals on the basepoint and both random sample sets.

Remark 8.8 (Level-uniformity and the tower). The five-stage proof structure relies only on (i) the Cayley–Dickson block decomposition, (ii) the composition-algebra norm identity at the preceding doubling level, and (iii) the anti-homomorphism property of conjugation. Each ingredient survives every doubling step provided x, y are taken in the imaginary part of the previous-level algebra. The identity is therefore not specific to $\mathbb{S}$: it is proved for every level in Theorem 9.3 of Section 9, which supplies the general argument. At the first higher level — the dim-32 pathion algebra $\mathbb{P} = \mathbb{S} \oplus \mathbb{S} \cdot \ell_{16}$ with $\ell_{16} = e_{16}$, and $p = (x, y) \in \mathbb{P}$ with $x, y \in \mathrm{Im}(\mathbb{S})$ — the identity reads $[R_{e_{16}}, G_p] = 0$, and the aphanic sublocus of $\mathbb{P}$ correspondingly carries an algebraic $U(1)$ center generated by the restriction of $R_{e_{16}}$ to the fiberwise kernel. The numerical values quoted at higher levels are certification witnesses for the implementation, not evidence for the identity.

9 The Cayley–Dickson Tower Recursion

The R_ℓ centrality identity of Section 8 was proven for the dim-16 case, with two ingredients used at the diagonal-block stage that depend on the composition-algebra structure of $\mathbb{O}$:

(i) $L_{\bar{X}} L_X = \|X\|^2 I$ on $\mathbb{O}$ for every $X \in \mathbb{O}$ (composition law);

(ii) $A = D = \|p\|^2 I$ in the diagonal blocks of $G_p = L_p^\top L_p$.

The composition law fails at the next CD level: $\mathbb{S}$ is not a composition algebra. Naively, this should obstruct the centrality proof at dim 32. In fact the obstruction is illusory: a strictly weaker identity, the *Imaginary L–R Symmetry Lemma*, suffices to drive the diagonal-block argument, and that identity is preserved at every CD level via flexibility alone. This section establishes the tower-wide recursion.

9.1 Imaginary Left–Right Symmetry

Theorem 9.1 (Imaginary L–R Symmetry)**.** *Let $\mathcal{A}_n$ be the 2^n-dimensional Cayley–Dickson algebra obtained by canonical recursive doubling. For every $n \geq 1$, every $X \in \mathrm{Im}(\mathcal{A}_n)$, and every $q \in \mathcal{A}_n$,*

$$\bar{X}(Xq) \;=\; (qX)\bar{X}. \tag{41}$$

Equivalently, $L_{\bar{X}} L_X = R_{\bar{X}} R_X$ as operators on $\mathcal{A}_n$.

Proof. Cayley–Dickson algebras are flexible at every level $\mathcal{A}_n$: $a(ba) = (ab)a$ for all $a, b \in \mathcal{A}_n$. Linearizing in a,

$$[a, b, c] \;=\; [c, b, a] \qquad \forall\, a, b, c \in \mathcal{A}_n,$$

the outer-pair symmetry of the associator. Apply this with $a = X$, $b = X$, $c = q$:

$$[X, X, q] \;=\; [q, X, X].$$

For $X \in \mathrm{Im}(\mathcal{A}_n)$, $X^2 = -\|X\|^2 \in \mathbb{R}$ commutes with everything in $\mathcal{A}_n$, so

$$X(Xq) - X^2 q \;=\; (qX)X - qX^2 \iff X(Xq) \;=\; (qX)X.$$

Negating both sides ($\bar{X} = -X$ for $X \in \mathrm{Im}$) gives $\bar{X}(Xq) = (qX)\bar{X}$, which is $L_{\bar{X}} L_X = R_{\bar{X}} R_X$. □

Remark 9.2 (The flexibility shadow of composition). At composition levels ($\mathcal{A}_n$ for $n \leq 3$, i.e., $\mathbb{C}, \mathbb{H}, \mathbb{O}$), both sides of (41) reduce independently to $\|X\|^2 q$, so each operator equals $\|X\|^2 I$. At $n \geq 4$ the operators each individually deviate from $\|X\|^2 I$ (reflecting the loss of composition), but they continue to agree with each other — their common value is some non-scalar operator that nonetheless inherits centrality with respect to the structures we construct.

9.2 Recursive R_ℓ Centrality

Theorem 9.3 (Recursive $R_{\ell_{n+1}}$ Centrality)**.** *Let $\mathcal{A}_{n+1} = \mathcal{A}_n \oplus \mathcal{A}_n \cdot \ell_{n+1}$, $n \geq 3$, and let $p = (X, Y) \in \mathcal{A}_{n+1}$ with $X, Y \in \mathrm{Im}(\mathcal{A}_n)$. Let $G_p^{(n+1)} := L_p^\top L_p$ on $\mathcal{A}_{n+1}$. Then*

$$[R_{\ell_{n+1}},\, G_p^{(n+1)}] \;=\; 0.$$

Proof. The block form of L_p on $\mathcal{A}_{n+1} \cong \mathcal{A}_n \oplus \mathcal{A}_n$,

$$L_p \;=\; \begin{pmatrix} L_X & -R_Y K_n \\ L_Y K_n & R_X \end{pmatrix}, \quad R_{\ell_{n+1}} \;=\; \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix},$$

where K_n is conjugation in $\mathcal{A}_n$, gives by direct 2×2 block multiplication

$$[R_{\ell_{n+1}},\, G_p^{(n+1)}] \;=\; \begin{pmatrix} -(B + B^\top) & A - D \\ A - D & B + B^\top \end{pmatrix},$$

where

$$A = L_{\bar{X}} L_X + K_n L_{\bar{Y}} L_Y K_n, \qquad D = R_{\bar{X}} R_X + K_n R_{\bar{Y}} R_Y K_n,$$

and B is a cross-block built from associator-action operators on $\mathcal{A}_n$.

Diagonal vanishing $A = D$. By Theorem 9.1 applied to X and to Y, $L_{\bar{X}}L_X = R_{\bar{X}}R_X$ and $L_{\bar{Y}}L_Y = R_{\bar{Y}}R_Y$. Substituting term-by-term, $A = D$.

Off-diagonal vanishing $B + B^\top = 0$. The argument is identical to that in Section 8 (Stages 3–5), using only the inner-product duality $\langle ab, c\rangle = \langle b, \bar{a}c\rangle$ (which holds at every CD level) and the conjugation anti-homomorphism property (also tower-universal). Both ingredients survive every doubling, so the cross-block argument ports verbatim to level $n+1$ for $X, Y \in \mathrm{Im}(\mathcal{A}_n)$.

Both diagonal and off-diagonal blocks vanish, so $[R_{\ell_{n+1}}, G_p^{(n+1)}] = 0$. □

Corollary 9.4 (Kernel-projector centrality at every level)**.** *At any level $n+1 \geq 4$ where $\mathcal{A}_{n+1}$ contains aphanic observers p (i.e., $\ker L_p \neq 0$) with halves in $\mathrm{Im}(\mathcal{A}_n)$, the orthogonal projector $\pi_p^{(n+1)}$ onto $\ker L_p$ commutes with $R_{\ell_{n+1}}$:*

$$[R_{\ell_{n+1}},\, \pi_p^{(n+1)}] \;=\; 0.$$

Proof. Spectral projectors are polynomials in $G_p^{(n+1)}$ via Lagrange interpolation across distinct eigenvalues; commutation with $G_p^{(n+1)}$ descends to commutation with each spectral projector. □

9.3 Empirical Confirmation at the Chingon Level

The proof of Theorem 9.3 relies on three identities that survive every Cayley–Dickson doubling: flexibility, inner-product duality, and the conjugation anti-homomorphism. None of these is special to the dim-32 case. The natural conjecture is that the recursion continues indefinitely.

Proposition 9.5 (Chingon-level certification probe, EMPIRICAL)**.** *At dim 64 (the chingon algebra $\mathbb{C}\approx$), the centrality identity*

$$[R_{e_{32}},\, G_p^{(64)}] \;=\; 0, \qquad p = (X, Y),\; X, Y \in \mathrm{Im}(\mathbb{P}),$$

is verified at machine precision across $N = 8$ random samples, with worst residual $3.05 \cdot 10^{-13}$. The structural argument of Theorem 9.3 ports verbatim to this level, so an algebraic proof is implied; the empirical verification serves as a sanity check.

The probe `op_chingon_centrality_sanity.cpp` computes the 64×64 Gram and right-multiplication matrices via the recursive CD product and certifies $\|[R_{e_{32}}, G_p^{(64)}]\|_F \leq 10^{-12}$. Build target included in the supplementary archive.

Remark 9.6 (Status of the tower recursion). Theorem 9.1 is PROVEN (algebraic). Theorem 9.3 is PROVEN for all $n \geq 3$: its proof uses only the inner-product duality and the conjugation anti-homomorphism property, both of which hold at every Cayley–Dickson level, so the argument is level-uniform and no doubling step is distinguished. Proposition 9.5 accordingly reports a corroborating certification probe at $n + 1 = 6$ (dim 64); it witnesses the implementation rather than carrying the claim, which the theorem already establishes at that level and above.

10 Open Frontiers

The framework as developed here closes the algebraic-foundations chapter of the Aphanes program. Several questions remain open as research directions; we list the principal ones, in order of proximity to the present results.

10.1 The $\mathrm{SU}(2)_L$–Fano Identification

Theorem 7.2 identifies the $\mathfrak{u}(1)$ factor of the holonomy algebraically as the restriction to $\mathcal{K}_{p_0}$ of the ambient operator R_ℓ. The corresponding identification of the $\mathfrak{su}(2)_L$ factor remains less direct: it is the restriction of the lifted SU(2)-stabilizer action, which is defined geometrically via the orbit structure rather than as a direct product of ambient operators. Numerical evidence indicates that the three generators G_2, G_3, G_4 of $\mathfrak{su}(2)_L$ correspond to right-multiplication by the imaginary quaternions $i, j, k \in \mathbb{H}$ acting on the rank-4 complement $\mathrm{Im}(Q)^\perp \cong \mathbb{H}$, where $Q = \mathrm{span}\{1, e_1, e_2, e_1e_2\}$ is the distinguished quaternionic subalgebra at the basepoint. An algebraic proof of this identification would parallel Theorem 8.1 for the abelian sister, closing the loop on the U(2) decomposition.

10.2 Universality of the Holonomy Classification at Higher Levels

Theorem 9.3 establishes the algebraic recursion of the U(1) center; the analogous question for the holonomy decomposition is open. At the pathion level (dim 32), aphanic observers exist (Two-Layer Theorem; (Cawagas, 2004) and probes), but the kernel structure is known to be richer: there is a rank-jumping kernel sheaf with strata of dimension 4 and dimension 0. Whether the canonical connection on the dim-4 stratum has $U(2)$ holonomy, deforms to a larger group, or ruptures at the next CD doubling is undetermined. The natural template, given Theorem 9.1, is to test whether the analogous stabilizer action on the dim-4 stratum produces a copy of $\mathfrak{su}(2)$ centralized by an ambient R_{ℓ_5}-restriction.

10.3 Global Integrability of the Kernel Distribution

Axiom 3 of the abstract framework (Skiouris, 2026) asserts non-integrability of the rank-4 kernel distribution $p \mapsto \ker L_p$ over the aphanic locus. This is supported empirically (no smoothly-defined four-form ω satisfying $d\omega \in I(\omega)$ has been found) but lacks a formal Frobenius test. A direct computation of the Frobenius bracket on the Gram–Schmidt-orthonormal kernel basis at a generic p would settle the matter. This is a finite computation in $\mathbb{O}$-arithmetic that has not yet been carried out.

10.4 Two-Layer Theorem Beyond Dim 32

Cawagas (Cawagas, 2004) conjectured, and we have empirically confirmed through dim 64, that fresh aphanic kernels (i.e., non-trivial $\ker L_p$ at points p that are not pulled back from lower levels) arise only at dim 16 and dim 32, vanishing thereafter. An algebraic proof of the Two-Layer Theorem at all higher levels remains open. The Schur complement argument at dim 32 indicates that the next obstruction to fresh kernels is a determinant condition that, empirically, becomes generic at dim 64+; a proof would generalize the keystone identity (15) to higher CD doublings.

10.5 Connection to Twistor Theory

The image of the kernel map $\Phi : \mathcal{V} \to \mathrm{Gr}(4, \mathbb{S})$ is $G_2/U(2)$, the twistor space of S^6 (Bryant, 1982). The algebraic structure of the kernel bundle pulled back from this twistor space, and its relationship to standard twistor constructions on S^6 (Bryant's nearly-Kähler structure, the canonical G_2-frame bundle), is a natural further line of inquiry — particularly given the U(2) holonomy and the fact that twistor spaces of nearly-Kähler 6-manifolds carry canonical Kähler structures with U(2) holonomy in their own right.

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